

What Substitutes for a Stale Mark? Breadth, Implied Volatility, and the Price of Being Late

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Abstract

A forecaster whose own data has gone stale must buy something in place of timeliness, and there is no accepted way to price that trade. Handicapping a forecaster of S&P 500 realised variance by δ days while every other source stays current, we define the substitution rate $R(\delta)$, the share of the delay's damage a candidate substitute repairs, and price two candidates against it over nineteen years.

The result is most legible in days. A fifty-five-day-old mark plus today's foreign index closes forecasts as well as a mark 4.6 days old [2.8, 7.7]; one day of staleness in that foreign feed costs 2.1 days [1.4, 3.3] of domestic freshness. Three conditions bound it. The rate is well determined only in stressed markets, where it is 81% [74, 87] at eleven weeks and identical in both halves of the sample; in calm markets so little damage is done that the ratio is undetermined. Against the strongest domestic control we could build it is 68% rather than 72%. And where the asset carries a liquid options market the cross-section is redundant, and that survives replacing thirty-day VIX with a horizon-matched nine-day index that forecasts better at every delay. That bounds the scope of the finding, not the method: quarterly-marked private equity, appraised property and lagged-NAV funds have no option chain to buy instead.

Two practical results sit inside the measurement. Fitting seven foreign series is the wrong way to buy breadth: one real-time factor, or an equal-weighted mean, does as well for a fraction of the estimation cost. And the trailing-median normalisation that puts eight markets on one scale hides the level of volatility from every model; restoring it is worth more at long delays than the entire cross-section.

1. The question

A private equity portfolio is marked once a quarter. Direct property is appraised on a cycle, and the resulting series is smoothed in ways that have been documented for thirty years [8]. Hedge fund returns arrive with a serial correlation that understates their true volatility [7]. In each case someone must produce a risk number today from marks that are weeks or months old, while public markets have gone on trading in plain sight.

The instinct is to reach for those markets. The question is which of them, and how far it gets you. Two candidates present themselves. The first is breadth: other markets are correlated with yours, and some of them have already closed today. The second is the options market, which reports a forward-looking risk number continuously and without any delay at all. These are different purchases, and a practitioner choosing between a wider data feed and an options subscription would like to know which one buys back more of what staleness took.

We cannot observe the truth for a private portfolio, so we run the experiment where the truth is observable. We take a forecaster of S&P 500 realised variance, withhold their own data for δ days, leave the other sources untouched, and measure what each gives back. For foreign equities the

asymmetry costs nothing to construct: Tokyo closes at 06:00 UTC, Hong Kong at 08:00, Mumbai at 10:00, London and Frankfurt at 16:30, New York at 21:00. On any given day the rest of the world has already finished trading when New York does.

2. What is already established

Four findings underpin this exercise, and none of them is ours.

Foreign equity volatility forecasts domestic volatility. Engle, Ito and Lin rejected the idea that volatility is a purely local phenomenon in 1990 [3], and the spillover literature since has measured the connections in considerable detail [2, 4]. Korkusuz augments a HAR model of S&P 500 volatility with a European block and improves on it [9]. Buncic and Gisler run the mirror experiment, forecasting seventeen foreign markets from US volatility, and report out-of-sample R^2 between 4.6% and 14.5% [1].

Non-synchronous closing times define which observations a forecaster may legitimately use. Zhao, Qiao and Lin say so explicitly and build an attention model whose mask prevents a closed exchange from acting as a source [15]. Jayawardena and co-authors forecast Tokyo volatility from markets that trade while Tokyo sleeps [6].

A live market can price a stale one. Goetzmann, Ivković and Rouwenhorst showed that same-day US returns predict the next day's international fund NAVs strongly enough to trade on [5]: the mechanism of this paper stripped to its simplest form.

Implied volatility carries much of the cross-market gain. This is the finding that shapes the design below. Both Korkusuz [9] and Buncic and Gisler [1] report that a large share of what looks like a cross-market effect is attributable to implied volatility, the VIX in particular. Their work does not settle the matter for a *delayed* forecaster, since in both cases the domestic information set is current, but it makes the comparison unavoidable: any claim that foreign markets substitute for timeliness has to be set against an options market that never goes stale.

What none of this work does is treat the delay as a variable. In every case additional data is added to an information set that is already current, and the question is whether it helps. Ours is the complementary question: when domestic information is not current, how much of the damage does each candidate undo, and how does that fraction move as the damage grows? That is a rate of substitution between things a practitioner can actually buy, and we could not find it measured.

3. How we measure it

The data are daily open, high, low and close for eight equity indices: S&P 500 (SPX), Nikkei 225 (N225), S&P/ASX 200 (AXJO), Hang Seng (HSI), Nifty 50 (NSEI), FTSE 100 (FTSE), DAX (DAX) and Bovespa (BVSP), from January 2000 to September 2026, together with three implied-volatility indices: CBOE VIX, its nine-day counterpart VIX9D, and VDAX-NEW. The two CBOE series are Cboe's own published daily histories; the eight indices and VDAX-NEW are retail end-of-day exports from Investing.com, downloaded in September 2026. The repository lists every file, the identifier it maps to, and the column dictionary, and pins the environment to NumPy and pandas alone. Each retail series came in two exports, because the source caps a download at 5,000 rows, which leaves nine joins to be checked rather than trusted. Eight of the nine overlap by 120 to 251 trading days, and across every overlapping row the two exports agree exactly: no mismatches, largest absolute

difference zero. The ninth is the S&P itself, where the exports abut without overlapping and no cross-check is possible. We report it as unverified and fall back to two weaker tests: the join is contiguous, Friday 9 March 2018 to Monday 12 March 2018 with no session missing or duplicated, and the return across it is -0.0013 , or 0.11 standard deviations of the series' own daily returns, so there is no level shift where the exports meet.

A foreign close is admissible only when its timestamp falls before the target's. Markets closing at or after the target therefore contribute their previous session. The rule generalises without special cases, which matters for Section 9: a Tokyo forecaster gets every other market a day late, because Tokyo closes first. It also disciplines the implied-volatility block, and the discipline is awkward in a way worth setting out precisely. VDAX-NEW satisfies the rule outright, since Frankfurt finishes hours before New York. VIX does not. Its printed daily close is stamped 21:15 UTC, fifteen minutes after the 21:00 S&P cash close, so that print is inadmissible. VIX is computed continuously from live SPX option quotes, so a value of it did exist at 21:00 and would be the right observation to use, but a retail end-of-day export carries no intraday snapshot and we do not have it. What we can run is a pair of timing specifications on either side of the one we want: the previous day's close, which is strictly admissible and discards a full day of information, and the same-day close, which contains fifteen minutes we are not entitled to. Section 8 leads with the lagged arm, because it is the one that obeys our own rule, and reports the same-day arm as a sensitivity check. The 21:00 observation is temporally between the two, and we are careful not to infer from that that its forecasting performance would be numerically between their results. Timestamps order; skill need not.

Volatility is the Yang-Zhang estimator [14] over rolling five-day windows, which uses the whole daily range and so sees more of each session than the close alone. Write O_s , H_s , L_s and C_s for the open, high, low and close of session s , and

$$o_s = \log(O_s / C_{s-1}), \quad c_s = \log(C_s / O_s), \quad u_s = \log(H_s / O_s), \quad d_s = \log(L_s / O_s)$$

for the overnight, open-to-close, high and low log returns. Over the n sessions ending at t the estimator is

$$\sigma^2_{YZ,t} = \sigma^2_o + k \sigma^2_c + (1 - k) \sigma^2_{RS}, \quad k = 0.34 / (1.34 + (n+1)/(n-1)), \quad (1)$$

where σ^2_o and σ^2_c are the sample variances of o_s and c_s over those n sessions, σ^2_{RS} is the mean over them of the Rogers-Satchell term $u_s(u_s - c_s) + d_s(d_s - c_s)$ [19], and $n = 5$ throughout. The three components measure the overnight gap, the drift within the session, and the range net of that drift; k is the weight for which the sum is minimum-variance and drift-independent under a Brownian price [14]. All variances use the $n - 1$ divisor.

Let i index the eight series and t the target's trading day. The decision variable each model sees is the log of that series' variance against its own recent level,

$$a_{i,t} = \log(\sigma^2_{YZ,i,t} / M_{i,t}), \quad M_{i,t} = \text{median}(\sigma^2_{YZ,i,s} : t - 251 \leq s \leq t), \quad (2)$$

and the quantity to be forecast, with horizon $h = 5$ trading days and target market denoted SPX, is

$$y_t = a_{\text{SPX}, t+h}. \quad (3)$$

So $M_{i,t}$ is the trailing 252-day median, and the implied-volatility series enter the same way, as the log of the level over its own trailing median. Dividing by a series' own median puts Tokyo, São Paulo and

the VIX on one scale and keeps each series' units out of the regression. One detail of that definition needs stating, because it looks like look-ahead and is not. The target a_{t+5} carries M_{t+5} , a median whose window includes the five days between the forecast origin and the outcome, which a forecaster at t does not know. No feature is affected, since every one is dated $t - \delta$ or earlier for the domestic block and t for the others. What changes is that the quantity being predicted is not one a practitioner could convert back to a variance forecast at the moment they make it. The implementable alternative normalises the outcome at the ORIGIN, $\log(\sigma^2_{YZ,t+5} / M_t)$, and we ran the whole experiment both ways. The two targets differ by the five-day drift in a 252-day median, which has a standard deviation 3.7% of the target's own and correlates +0.52 with it, so it is neither large nor independent. It changes nothing: the substitution rate is 71% against 72% at eleven weeks, and no delay differs by more than a point. We keep the target-dated definition and report the origin-dated run as the check that licenses it. The target is a_{t+5} , five trading days ahead, left continuous: a companion note [13] showed that splitting such a target at its median throws away exactly the magnitude information a cross-sectional model would use, so we do not split it.

Models are built from three feature blocks. The domestic block is the target's own history at daily, weekly and monthly aggregation, in the manner of Corsi [2], observed at $t - \delta$. The foreign block is seven index decision variables at their admissible timestamps: the six that close before New York, plus Bovespa a session late. The implied-volatility block is VIX and VDAX. Only the domestic block is delayed; the others are what a forecaster could see today, and that asymmetry is the whole point of the comparison. Every model is a ridge regression refitted every 21 days on a rolling 1,250-day window, with the penalty chosen on a 250-day validation tail and features standardised on training statistics alone. Formally, let x_t be the feature vector a model sees at t , assembled from the blocks above, and let $T(t)$ be the 1,250 days of the training window in force at t . Features are centred and scaled by that window's own mean μ_T and standard deviation s_T , giving $z_t = (x_t - \mu_T) / s_T$ elementwise, and the coefficients (β_0, β) solve

$$\min_{\beta_0, \beta} \sum_{u \in T(t)} (y_u - \beta_0 - z_u' \beta)^2 + \lambda \beta' \beta, \quad (4)$$

with the intercept β_0 left unpenalised and λ chosen from $\{0.3, 1, 3, 10, 30\}$ on the last 250 days of the training window, which are held out of the fit that selects it. The forecast is $f_t = \beta_0 + z_t' \beta$. Standardising on training statistics alone matters: scaling a feature by a full-sample standard deviation would let the test period influence the fit through the scale, which is a small leak but a real one. All are trained on the same labels, a pair being admitted only once its outcome was resolvable at $t - \delta$, so differences between them are features and nothing else. The rolling fixed window is not incidental; it is what makes the tests in Sections 6 and 7 valid.

Losses need care here. With an imperfect volatility proxy only a restricted family of losses preserves the ranking a researcher would obtain against true latent volatility [17], and that family is defined on the VARIANCE scale. Our target is a log variance, so the two halves of the usual pairing land differently. QLIKE survives the transformation: it depends on the ratio of outcome to forecast alone, and dividing both by the same trailing median leaves that ratio untouched, so what we compute is exactly the variance-scale QLIKE. Squared error does not. Computed on log variance it is outside the family, because a monotone transform of the arguments is precisely what the robustness result fails to survive.

We therefore report three. Let m be the training-window mean of y , the benchmark forecast; let $v_t = \exp(y_t) M_t$ be realised variance on its natural scale and $w_t = \exp(f_t) M_t$ the corresponding variance forecast; and let n be the number of test days. The three losses are

$$R^2 = 1 - \sum_t (y_t - f_t)^2 \div \sum_t (y_t - m)^2, \quad (5)$$

$$\text{QLIKE} = (1/n) \sum_t [v_t/w_t - \log(v_t/w_t) - 1], \quad (6)$$

$$\text{MSE}_{\text{var}} = (1/n) \sum_t (v_t - \gamma_t w_t)^2, \quad \gamma_t = (1 / |T(t)|) \sum_{u \in T(t)} \exp(y_u - f_u), \quad (7)$$

where γ_t is Duan's smearing factor [20], estimated from training-window residuals at each refit. Skill under QLIKE and MSE_{var} is defined as in (5), one minus the model's loss over the benchmark's, so all three columns are shares of a benchmark loss removed and are read the same way. Of the three, R^2 in (5) is the paper's headline and is *not* proxy-robust; QLIKE in (6) is; and MSE_{var} in (7) is Patton's MSE on the variance scale and is also robust. The smearing factor is there because $\exp()$ of a log forecast is median-unbiased, not mean-unbiased, and squared error on the variance scale punishes exactly that, so the reconstruction needs the correction, not the raw exponential. Mean absolute error, used in some recent work in this area, is in no version of the family. One choice inside that arrangement needs defending, because it looks backwards: the headline number in this paper is the one loss of the three that is NOT proxy-robust. We lead with log-scale R^2 because the paper's object is a ratio, $R(\delta)$, and a ratio needs a denominator a reader can hold: the share of a delay-induced loss recovered, naturally bounded near 0 and 100%, on the same scale as the companion note [13] and as the decision variable itself. QLIKE skill on a variance scale is the right loss and the wrong ruler for that purpose; its levels are not comparable to anything else in this literature. The defence is that the choice is presentational and is made where it can be checked: both proxy-robust losses are reported beside the headline in every table where a conclusion turns on one, and they support the same conclusions everywhere except the $\delta = 0$ row, where they disagree with each other as well and Section 9 withdraws the claim instead of picking a winner. A reader who prefers the robust family throughout can read the QLIKE columns and will reach the same conclusions with different numbers attached. Where conclusions differ across losses we give weight to what both proxy-robust losses support; where those two disagree with each other, we treat the finding as unresolved, and we say so at the one place in the paper where that happens.

Write $S(\delta)$ for skill at delay δ under any one of (5), (6) or (7): the share of the benchmark's loss that a model removes. $S_{\text{own}}(\delta)$ is the skill of the model holding only the domestic block, δ days stale; $S_{\text{alt}}(\delta)$ is the skill of the same model with a candidate substitute added. Delay costs $S_{\text{own}}(0) - S_{\text{own}}(\delta)$; the substitute returns $S_{\text{alt}}(\delta) - S_{\text{own}}(\delta)$. The substitution rate is their ratio,

$$R(\delta) = [S_{\text{alt}}(\delta) - S_{\text{own}}(\delta)] \div [S_{\text{own}}(0) - S_{\text{own}}(\delta)]. \quad (8)$$

It has no units, so it can be compared across loss functions, across markets, and across candidates. One caution, which Section 8 will need: the measure assumes the substitute is *restoring* something delay destroyed. A source that also carries information the domestic history never had can exceed 100%, and when it does, the ratio has stopped answering the question it was built for. The denominator can fail in the other direction too: where delay costs a forecaster little, $S_{\text{own}}(0) - S_{\text{own}}(\delta)$ approaches zero and the ratio need not have a bounded confidence set. The Appendix gives a construction that stays honest there, and Section 10.2 applies it to the state where the risk is

greatest. On this sample the denominator never gets close enough for the pathology to bite, but it is a property of the measure itself, and anyone reusing $R(\delta)$ on a series where delay does little damage should check it rather than assume it.

4. What breadth recovers

The test window runs from May 2007 to September 2026, 4,862 days.

Table 1. Out-of-sample skill against the training mean, and the substitution rate under each loss. Skill is the fraction of benchmark loss removed, so the two pairs are comparable. Intervals bootstrap the whole ratio, recomputing numerator and denominator on the same resampled days. R^2 is on the log scale and is not proxy-robust; QLIKE is.

δ	R^2 own	R^2 cross	$R(\delta)$ on R^2	QLIKE skill own	QLIKE skill cross	$R(\delta)$ on QLIKE
0	0.4987	0.4884	n/a	0.5721	0.5578	n/a
1	0.4518	0.4512	−1% [−32, 26]	0.5196	0.5173	−4% [−43, 31]
3	0.3848	0.4262	36% [17, 54]	0.4384	0.4920	40% [16, 62]
5	0.3371	0.4202	51% [36, 66]	0.3738	0.4855	56% [37, 73]
8	0.2822	0.4111	60% [47, 71]	0.2873	0.4749	66% [50, 78]
13	0.2011	0.3907	64% [53, 73]	0.1309	0.4526	73% [59, 83]
21	0.1121	0.3726	67% [57, 76]	−0.0182	0.4358	77% [64, 86]
34	0.0404	0.3603	70% [61, 77]	−0.1366	0.4282	80% [69, 87]
55	−0.0351	0.3470	72% [64, 79]	−0.1769	0.4223	80% [69, 88]

Read the table down a column and then across. The first column is what a forecaster loses to delay alone: R^2 on the domestic block falls from 0.4987 at zero delay to −0.0351 at eleven weeks, so by then the stale mark is worse than quoting the training mean. The second column is the same forecaster with the foreign cross-section added, and it barely moves, from 0.4884 to 0.3470, because the peers are current whatever the domestic delay is. The third column is the ratio of the two gaps, which is $R(\delta)$ in (8): 36% of the delay's damage repaired at three days, 51% at a week, 72% at eleven weeks. The two QLIKE columns repeat the exercise on the proxy-robust loss and reach the same shape with larger numbers, 40% to 80%, so the result is not an artefact of the scale the headline is quoted on. The $\delta = 1$ row is the one that does not fit the pattern, and it is the row that should not: at one day the domestic history is barely stale and there is almost nothing for the cross-section to put back, so the rate is indistinguishable from zero under both losses.

One warning attaches to every figure in that third column before the paper goes further, and Section 10.2 establishes it. An unconditional rate averages two states in which this mechanism behaves quite differently. In stressed markets it is 81% at eleven weeks with an interval of [74, 87], and it is 81% in each half of the sample taken separately: a well-determined, stable number. In calm markets the same quantity spans ninety points and cannot be pinned down at all, for a mechanical reason: where delay costs a forecaster almost nothing, the share of that cost repaired by anything has no stable value. Table 1's 72% is the average of a result and a non-result. We report it because it is the prespecified estimate and because the literature quotes unconditional rates, but the number a practitioner should carry is the conditional one in Table 17.

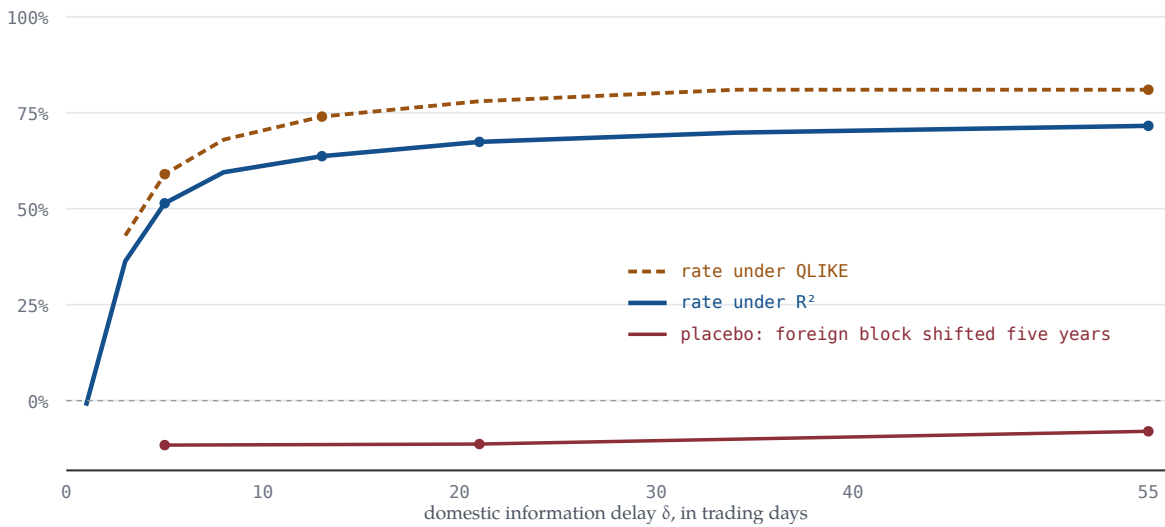


Figure 1. Breadth buys back almost nothing at one day, half the loss within a week, and around three-quarters by eleven weeks. The placebo, described in Section 7, keeps the same foreign series but destroys their alignment with the target, and the effect disappears.

Three things stand out. At one day the rate is indistinguishable from zero, as it should be: the S&P's own recent history is still the best available description of the S&P, and the foreign block we selected produces no measurable net improvement on it. By eleven weeks the picture has inverted. The domestic model's out-of-sample R^2 has gone negative, meaning it would be better to forecast the training mean, while the cross-sectional model still holds 0.347, about 70% of what was available with fully current data. And between those extremes the rate climbs steeply and then slows, which raises the obvious question of where it stops.

Section 3 promised a third loss, and it belongs here because it is the robust one that most resembles the headline. Under squared error on the variance scale the substitution rate reads 85% at five days and 90% at eleven weeks, against 51% and 72% on the log scale. Both robust losses therefore put the rate ABOVE the headline, which makes the number we lead with the conservative one. The variance-scale intervals are very wide at short delays, $[-74\%, 126\%]$ at five days, because the denominator there is small and squared error on raw variance is dominated by a handful of crisis days; that is why QLIKE, not MSE, is the robust loss we lean on.

5. Why breadth stops where it does

The rate flattens near three-quarters instead of approaching one, and the reason is not subtle once measured. At very large delay the domestic model retains little usable out-of-sample skill, so the cross-sectional model is working almost entirely from foreign information. Its ceiling is therefore whatever share of the S&P's volatility the rest of the world can see.

We measured that share directly. Regressing the S&P's decision variable on the six foreign closes of the same day gives $R^2 = 0.695$. Roughly 30% of the S&P's current volatility state is not linearly spanned by those six series in this sample. We resist calling that share genuinely local, because the regression does not show it: a non-linear function of the same closes, or a wider cross-section, could reach some of it. What the number does bound is how much a LINEAR model of THESE markets can see, the ceiling the fitted model is working against. Predicting five days ahead from foreign closes alone gives $R^2 = 0.399$ against 0.498 from domestic history, or 80% of it, which brackets the 72% the

fitted model achieves. The gap between the in-sample bound and the fitted figure is consistent with the price of estimating parameters instead of knowing them, and Section 5 measures a cost of the right order; the design does not establish that estimation is the whole of it.

The cross-section is also nearly one-dimensional. The first principal component of the six foreign volatility series accounts for 66.5% of their variation and the first two for 76.9%. Six markets are close to one global volatility factor plus idiosyncratic noise, which carries a practical implication: an eighth or a ninth correlated index may add relatively little under this linear specification, because what a linear model of these markets can see is set by how much of the S&P is global; counting more global series does not move it.

That leaves the other half of the sentence above: what it costs to estimate parameters instead of knowing them. We had treated this as a residual, and it is large enough to deserve a number. Breadth is not free. Seven foreign coefficients must be estimated from a finite window, and the estimation is paid for whether or not the coefficients turn out to matter.

Two measurements agree on the size of the bill. The first varies the number of peers at $\delta = 0$ and reads off the slope: the penalty is a straight line through the origin at -0.0014 of R^2 per market, which accounts for 98% of the variation across peer counts and implies -0.0100 for the full seven. The second replaces the foreign block with AR(1) surrogates fitted to each peer, drawing innovations from a multivariate normal with the peers' own contemporaneous covariance so the surrogate cross-section is as correlated as the real one, while carrying no information about the target at any lag. Averaged over one hundred draws this gives -0.0131 at $\delta = 0$, with a Monte Carlo standard error of 0.0003 , an order of magnitude below the sampling interval that follows and therefore not the uncertainty that binds. The two methods differ by 0.003 , which agrees on the magnitude without being a formal match, and we do not read a third decimal place from either. About one point of R^2 is the price of carrying the cross-section. Section 5.2 measures it a third time, with a generator that fits nothing at all, and gets the same answer.

Two things follow. The first settles a reading we had left open. At $\delta = 0$ the measured difference is -0.0103 and the cost is -0.0131 , so the gross content of the foreign block is $+0.0028$. Gross is a difference of two estimated quantities, so we bootstrap it, resampling test days in moving blocks and recomputing the measured difference and the surrogate cost on the same days so that their errors stay correlated. The interval is $[-0.0060, +0.0115]$: indistinguishable from zero, and bounded above at about one point of R^2 . Foreign markets are not misleading about the current state of the S&P. Whatever they know about it that fully current domestic data does not already carry is too small for this design to detect, and the negative sign was the bill, not the information.

The second is that the bill grows with delay, from -0.013 at $\delta = 0$ to -0.054 at $\delta = 21$. This is a property of the fitting procedure: as domestic information goes stale the fit weakens, the validation step selects a looser penalty, and the seven regressors are given more rope. Separating the two makes the gross and net readings diverge in a way that matters for anyone deciding whether to build this.

Table 2. What the cross-section knows, and what a forecaster keeps after paying to estimate it. Cost from AR(1) surrogates, one hundred draws, innovations carrying the peers' own cross-market covariance. Gross is bootstrapped in the text: indistinguishable from zero at $\delta = 0$, marginal at $\delta = 1$, positive from $\delta = 2$.

δ	0	1	2	3	5	8	13	21
Net ΔR^2 (measured)	-0.0103	-0.0006	+0.0188	+0.0414	+0.0831	+0.1289	+0.1896	+0.2605
Cost of breadth	-0.0131	-0.0155	-0.0180	-0.0201	-0.0244	-0.0306	-0.0409	-0.0536
Gross ΔR^2	+0.0028	+0.0149	+0.0368	+0.0616	+0.1075	+0.1595	+0.2305	+0.3141

Gross content clears zero at two days, $[+0.0179, +0.0555]$, and the net figure crosses there too. At one day the gross interval is $[+0.0009, +0.0280]$, which excludes zero by a margin of 0.0009, close enough to the boundary that we report it and decline to build on it. An earlier draft, using ten surrogate draws instead of a hundred, put that interval across zero and we withdrew the reading entirely; a better cost estimate moved it back. A band in which the foreign block holds something real that cannot yet pay for its own estimation is therefore consistent with these data at one day, and no more than that. At $\delta = 21$ estimation consumes 17% of what the cross-section knows, +0.3141 gross against +0.2605 net. That gap is not a law of nature. It is the cost of a particular estimator, and a cheaper one should recover part of it: a factor representation of the foreign block instead of seven free coefficients, which the principal-component result above already suggests would lose little. We have not built that, and we make no claim about how much it would recover.

5.1 Fitting the cheaper representation, instead of inferring it

Two facts above sit next to each other without being joined. The foreign block is nearly one-dimensional, its first principal component accounting for 66.5% of the variation and its first two for 76.9%. Carrying seven regressors costs about a point of R^2 at $\delta = 0$ and more as the delay grows, measured by replacing the block with surrogates that carry no signal. It is natural to conclude that a cheaper parameterisation would recover part of that cost. That is an inference from a hypothetical generator, not a demonstration, and the demonstration is cheap.

So compress the block and compare. The eigenvectors are recomputed at every refit from the covariance of the training window alone, on columns standardised with training statistics alone, with each eigenvector's sign pinned so the factor does not flip arbitrarily between refits. A basis computed once on the whole sample would leak: "the first component of the foreign block" is not something a forecaster in 2009 could have written down from 2026 data. An equal-weighted mean of the seven series is included as the compression that requires no estimation at all.

Table 3. Out-of-sample R^2 for four representations of the same seven foreign series, against the domestic-only model. PC1 and PC2 use one and two real-time global factors; FULL is the seven-coefficient model the rest of the paper uses. The last columns test PC1 against FULL, which are non-nested, by Giacomini-White.

δ	domestic	+ mean	+ PC1	+ PC2	+ all seven	PC1 – all seven	GW z
0	0.4987	0.4999	0.5001	0.4975	0.4884	+0.0117	2.91
3	0.3848	0.4374	0.4385	0.4358	0.4262	+0.0124	1.95
5	0.3371	0.4310	0.4325	0.4300	0.4202	+0.0123	1.87
13	0.2011	0.4005	0.4032	0.4012	0.3907	+0.0125	1.74
21	0.1121	0.3841	0.3873	0.3842	0.3726	+0.0147	1.87
55	−0.0351	0.3617	0.3664	0.3630	0.3470	+0.0194	2.08

One real-time factor beats all seven series at every delay, by 0.012 to 0.019 of R^2 , and the margin grows with delay exactly as the surrogate cost does. The gain reaches conventional significance at $\delta = 0$ and $\delta = 55$ and falls short of it in between, so the pattern is more convincing than any single entry. The real-time first component explains 66.6% of the block's variance averaged over refits, against 66.5% in the full-sample figure quoted above, which says the basis is stable enough to be estimated as you go.

Reporting a new preferred specification on the log scale alone would be inconsistent with Section 7, which exists to establish that the scale changes the answer. Under QLIKE, the proxy-robust loss, one factor wins at every delay, exactly as it does under R^2 . Under squared error on the variance scale a compression wins at five of six delays; the exception is $\delta = 0$, where the seven-regressor model scores lowest. That loss is dominated by a handful of crisis days, so Section 7 reports its intervals as uninformative at short delays, so the exception is a statement about tails. Where the three losses appear to disagree elsewhere they are choosing between one factor and two, which differ by thousandths and are not a disagreement about anything.

The compression also raises the substitution rate, and a referee has asked which number the paper is claiming. Recomputed with the same full-ratio bootstrap as the headline, $R(55)$ is 71.6% [63.6%, 78.8%] for the seven-regressor model, 74.3% [67.2%, 80.9%] for the equal-weighted mean and 75.2% [68.2%, 81.8%] for one factor. Every compressed estimate sits inside the interval the seven-regressor model already carries, so the two are not distinguishable and we do not replace the headline with the larger figure. We report 72% as the prespecified full-block estimate, note the compressed estimate beside it, and say plainly that the difference is inside the noise of both.

Two qualifications, and the first is the more important. The equal-weighted mean captures nearly all of the gain: PC1 beats it, but by 0.001 to 0.005. What is being bought is the decision to compress, not the machinery of compressing, and a practitioner should take the average and skip the eigenvector. Second, this changes the $\delta = 0$ row. The seven-coefficient model scores 0.4884 against the domestic model's 0.4987, the small negative this paper has declined to interpret, while PC1 scores 0.5001, above it. We do not claim the cross-section helps at $\delta = 0$ on that basis: it is a difference of 0.0014 with no interval attached, and reading an uninterval-ed difference as a finding is the specific error this paper has made and corrected three times. What the row supports is narrower and sufficient, that the negative at $\delta = 0$ belongs to the parameterisation and not to the information.

5.2 Is the bill an artefact of the generator?

Both measurements of the bill are relative to something. The slope method extrapolates from peer counts; the surrogate method replaces the foreign block with fitted Gaussian AR(1) series. A referee objects to the second, correctly, that it preserves first-order persistence and contemporaneous cross-market covariance and nothing else: not heavy tails, not volatility-of-volatility, not crisis clustering, not higher-order serial structure. Under that objection the cost is identified only relative to a generator we chose, and a forecaster's real regressors are not Gaussian AR(1) series.

The objection has a direct test. Measure the bill again with a generator that fits nothing at all: take the real foreign block and move it circularly through the sample. Every marginal, every tail, every cluster is then the real one, and the only thing destroyed is the alignment with the target. If an unfitted generator returns the fitted generator's number, the objection is answered, not conceded.

Table 4. The cost of breadth under a fitted and an unfitted null, on identical days. AR(1) is the generator used above, twenty draws; SHIFT is the real foreign block moved circularly with its real missing-data mask held fixed, twenty offsets, with the Monte Carlo standard error of that mean beside it. NAIVE is the same shift applied to the raw array, so the holidays move with the values; it is reported as a diagnostic, not as a candidate.

δ	net ΔR^2	cost, AR(1)	cost, SHIFT	(s.e.)	cost, NAIVE	gross, AR(1)	gross, SHIFT
0	-0.0103	-0.0125	-0.0122	0.0009	-0.0329	+0.0021	+0.0019
1	-0.0006	-0.0148	-0.0140	0.0010	-0.0331	+0.0142	+0.0134
3	+0.0414	-0.0198	-0.0176	0.0013	-0.0342	+0.0612	+0.0590
5	+0.0831	-0.0242	-0.0209	0.0016	-0.0355	+0.1073	+0.1040
13	+0.1896	-0.0409	-0.0349	0.0027	-0.0436	+0.2306	+0.2245
21	+0.2605	-0.0546	-0.0487	0.0039	-0.0519	+0.3151	+0.3092
55	+0.3821	-0.0701	-0.0694	0.0071	-0.0683	+0.4522	+0.4515

One bookkeeping point before the result, because the table invites a fair objection. The AR(1) column here is twenty draws on this subsection's delay grid, not the hundred behind Table 2, so the two tables give slightly different figures for the same quantity: -0.0125 against -0.0131 at $\delta = 0$, and -0.0546 against -0.0536 at $\delta = 21$. That spread is Monte Carlo error and nothing else. Table 2 remains the paper's estimate of the bill, because it rests on more draws. What Table 4 is for is the comparison *within* it, between two columns computed on identical days from an identical number of draws, and that comparison is unaffected by how many draws either column uses.

The two generators agree at every delay. Compared against three times its own combined Monte Carlo standard error, delay by delay rather than against the loosest tolerance in the table, the largest gap is 0.0060 of R^2 at $\delta = 13$, which is 0.60 of the tolerance there, and no delay comes closer. The gross readings move by less: +0.0021 becomes +0.0019 at $\delta = 0$, and +0.3151 becomes +0.3092 at $\delta = 21$. Nothing in Section 5 depends on which generator supplied the number, and in that sense the referee's objection is answered. That claim is worth only as much as the null generator behind it, and an earlier version of this measurement was wrong twice before it was right. Appendix B records both mistakes and what they cost, because a placebo that quietly stops being a placebo returns a plausible number with no outward sign of trouble.

Getting that generator right took two attempts and both failures are worth a line, because each returned a plausible number rather than an error. The first shifted the block without preserving its

pattern of missing observations, which quietly discarded about 187 test days per draw - 3.8% - from every comparison. The second drew shift offsets without a guard against the ends, so one draw landed eight days from the identity, scored +0.1893 on its own against a spread with a standard deviation of 0.0709, and dragged an eight-draw mean. Corrected, the naive version of the penalty sits 0.0207 of R^2 below the corrected estimate at $\delta = 0$ and 0.0011 below it at $\delta = 55$. A placebo that quietly stops being a placebo returns a believable figure with no outward sign of trouble, so the shift is now guarded and the mask preserved.

5.3 Is breadth a global factor by another name?

A referee raises omitted-variable bias: foreign breadth may not be capturing cross-border diffusion at all, but proxying for a global systemic risk factor, so that what looks like transmission is simultaneous reaction to shared news. Section 5.1 is most of the way to agreeing, since one real-time factor beats all seven series at every delay, but it never separates the two halves and asks which one carries the recovery. The decomposition is available at no cost, because the basis is already being estimated.

At each refit, write v_1 for the leading eigenvector of the training covariance of the standardised foreign block. Let p_t be that block's seven decision variables at t , standardised on the training window, and let v_1 be the unit eigenvector belonging to the largest eigenvalue of their training covariance, with its sign pinned so that the first loading is positive. The split is

$$p_t = g_t v_1 + r_t, \quad g_t = p_t' v_1, \quad r_t = p_t - g_t v_1, \quad (9)$$

where g_t is the scalar global factor and r_t is the six-dimensional remainder orthogonal to v_1 . Sign pinning is not cosmetic: an eigenvector is defined only up to sign, and one that flipped between refits would enter the regression as noise unrelated to the data. The + *PC1* column below uses g_t as a single regressor and + *remainder* uses the six components of r_t . Each day's standardised row x splits into a global part $(x \cdot v_1) v_1$ and a remainder orthogonal to it, spanned by components two through seven. Adding the two back together is a rotation of the raw block and must reproduce it: it does, to within 0.0010 of R^2 at every delay, the check that licenses reading the split.

Table 5. The foreign block split into its global component and the remainder orthogonal to it, both estimated in real time at every refit. The last column tests PC1 against the remainder by Giacomini-White; positive favours the global component.

δ	domestic	+ PC1	+ remainder	+ all seven	PC1 – domestic	remainder – domestic	GW z
0	0.4987	0.5001	0.4873	0.4884	+0.0014	−0.0114	2.94
3	0.3848	0.4385	0.3737	0.4262	+0.0538	−0.0111	4.92
5	0.3371	0.4325	0.3276	0.4202	+0.0954	−0.0095	5.60
13	0.2011	0.4032	0.1943	0.3907	+0.2021	−0.0067	6.01
21	0.1121	0.3873	0.0992	0.3726	+0.2753	−0.0128	5.94
55	−0.0351	0.3664	−0.0629	0.3470	+0.4015	−0.0277	7.22

The answer is unambiguous. The global component alone beats the domestic model at five of six delays and by as much as 0.40 of R^2 . The remainder, six real-time components orthogonal to it, is *negative* at every delay: it beats the domestic model at none, and carrying it costs between 0.007 and

0.028 of R^2 , the estimation bill of Section 5 with no information to pay for it. Giacomini-White separates the two at every delay, from $z = 2.94$ to 7.22 .

So the referee is right about the mechanism and we adopt the correction. What this paper measures is a global volatility factor observed while the domestic series is stale, not bilateral diffusion between particular markets. Three things follow. The substitution rate is unchanged: it does not matter to the number what the factor is called. The practical advice sharpens: a forecaster wants a cross-market average, not a panel of countries, as Section 5.1 already recommended for a different reason. And a limitation appears that Section 11 now carries: a first principal component of seven equity indices is a proxy for global risk, not a measurement of it, so orthogonalising against it is weaker than orthogonalising against a factor built from a wider universe, and the remainder may still hold global content this basis cannot see.

6. A model of the delay curve

Everything above is a measurement. Two of those measurements are close to a specification, and writing it down turns the curve in Figure 1 from a description into a prediction with a testable ceiling.

Section 5.3 found the recovery entirely in the leading component of the foreign block, with the six orthogonal components worth less than nothing. Section 10.1 finds that the level of volatility, which equation (2) removes, carries information that grows with delay. Together they say: log variance is a slow level, plus a factor several markets share, plus something idiosyncratic. For market i ,

$$\log \sigma_{i,t}^2 = \mu_{i,t} + \lambda_i g_t + e_{i,t}, \quad g_t = \varphi g_{t-1} + u_t \quad (10)$$

with μ the slow level the 252-day median tracks, g a persistent global factor, e idiosyncratic and u white noise. The decision variable of equation (2) subtracts an estimate of μ , so $a_{i,t} \approx \lambda_i g_t + e_{i,t}$. That is the whole of Section 10.1 in one line: the normalisation removes μ by construction, so restoring it is worth something.

A forecaster δ days late sees $a_{SPX,t-\delta}$, one noisy reading of $g_{t-\delta}$. The cross-section supplies seven readings of g_t , whose common component suppresses the idiosyncratic part. Under (10),

$$E[g_{t+h} | g_{t-\delta}] = \varphi^{h+\delta} g_{t-\delta}, \quad E[g_{t+h} | g_t] = \varphi^h g_t \quad (11)$$

and three consequences follow, each checkable against quantities measured elsewhere in this paper, not fitted to the curve.

First, the stale forecaster's skill decays geometrically: $\log S_{\text{own}}(\delta)$ is linear in δ with slope $2 \log \varphi$. Fitted over the eight delays where S_{own} is positive, the line has $R^2 = 0.9984$ and slope -0.07217 per day, implying $\varphi = 0.9646$. The factor's own persistence, estimated from the principal-component series and never shown the decay curve, is 0.9678 . The two agree to 0.0033 . The delay curve is the factor's memory, seen through a forecast.

Second, $S_{\text{cross}}(\delta)$ should be near-flat, because the foreign block delivers g_t whatever the domestic delay is. Across eleven weeks it falls 29%, against 107% for the domestic model. Perfect flatness is not predicted, because the domestic block is still in the cross-sectional model and still decays, but the contrast is the model's.

Third, and most useful: $R(\delta)$ rises from zero towards a ceiling below one, and the model names it. Since $S_{\text{own}}(\delta) \rightarrow 0$,

$$R(\infty) = S_{\text{cross}} \div S_{\text{own}}(0), \quad (12)$$

which is below one because seven peers estimate g_t with error and never observe $e_{\text{SPX},t}$. Both quantities are measured elsewhere: $S_{\text{cross}}(55) = 0.3470$ and $S_{\text{own}}(0) = 0.4987$, giving a predicted ceiling of 69.6%. The measured curve reaches 69.8% at $\delta = 34$ and 71.6% at $\delta = 55$. Nothing in that prediction was fitted to the curve it predicts.

So the substitution rate is not an arbitrary rising function. It is a factor share, it converges, and the paper can say what it converges to. The model also explains why Section 5.3's orthogonal remainder is worthless, being e , which forecasts nothing, and why Section 9.3's staleness penalty exists, since an aged foreign block reads g at $t - g$, not at t and (11) prices that at φ per day. Three results the paper had separately are one mechanism.

What the model does not explain is why the mechanism is stronger in stress. Equation (11) holds λ and φ constant; Section 10.2 shows the factor doing more work when markets are frightened, which would need λ or the variance of u to move with state. We have not fitted that, and it is the obvious next specification.

7. Does it survive testing

Bootstrap intervals describe a difference; they do not test a hypothesis, and our domestic model is nested inside our cross-sectional one, which rules out the usual comparison. Diebold has written at length about the misuse of his own test in exactly this situation [12]. We use Giacomini-White [10] as the primary test, because it is valid for nested models and, being loss-agnostic, covers the QLIKE result as well as the squared-error one. It requires a fixed rolling estimation window, of the kind Section 3 describes. Writing L_t^{own} and L_t^{alt} for the two models' losses on test day t and $D_t = L_t^{\text{own}} - L_t^{\text{alt}}$ for the loss differential, the statistic is

$$GW = d \div SE(d), \quad SE(d)^2 = (1/n) \left[\gamma_0 + 2 \sum_{j=1}^L (1 - j/(L+1)) \gamma_j \right], \quad (13)$$

where d is the mean of D_t over the n test days, γ_j is its sample autocovariance at lag j , and the Bartlett weights $(1 - j/(L+1))$ run to $L = 9$ lags. The HAC correction is not optional here: a five-day-ahead target overlaps on four of every five days, so the raw standard error of a daily loss differential understates by roughly a factor of two. Positive GW favours the model with the substitute, and we report two-sided p -values from the standard normal throughout. Clark-West [11] serves as a second opinion on squared loss, with a qualification developed in Section 7.1 that is the reason it is the second opinion, not the first.

Table 6. Test statistics, HAC-adjusted with a Bartlett kernel at nine lags. Positive favours the cross-sectional model. All entries have two-sided $p \leq 0.011$; all except QLIKE at $\delta = 3$ have $p < 0.01$.

δ	3	5	8	13	21	34	55
Giacomini-White, squared loss	3.18	4.48	5.08	5.34	5.66	6.21	7.02
Giacomini-White, QLIKE	2.57	3.18	3.24	2.77	2.87	3.00	3.28
Clark-West	7.09	8.02	8.28	8.09	8.01	8.51	9.21

P-values are two-sided throughout. On that basis every statistic from $\delta = 3$ onward has $p \leq 0.011$, and all but one are below 0.01: the exception is QLIKE at $\delta = 3$, where $z = 2.57$ gives $p = 0.0102$. A one-sided test against the directional alternative that the cross-sectional model is better would put every entry under 0.01, and the alternative is directional in substance, but we report the two-sided figure because it is the more demanding of the two. QLIKE, the harder loss to move, is significant at conventional levels throughout.

7.1 A caveat on the Clark-West column

Clark-West is the second opinion on squared loss and carries a qualification worth stating. Its correction term assumes the larger model nests the smaller exactly; a ridge with a cross-validated penalty satisfies that only approximately, because shrinkage pulls the larger model away from the nesting, and the correction can then flatter it. The diagnostic below measures how far from exact the nesting is at each delay, and the remedy re-runs the statistic on an unpenalised fit, where nesting is exact by construction. The diagnostic is a ratio with no established critical value. We offer that ratio as a description, not a threshold, and lean on the unpenalised re-run instead of on where the ratio happens to fall.

Table 7. A diagnostic and a remedy for the Clark-West column, applied to this paper's own statistics. The ratio is the mean Clark-West adjustment over the magnitude of the mean loss differential; the common-penalty columns impose a single ridge penalty, selected on the domestic arm, across both.

δ	refits where the penalties differ	adjustment ratio	Clark-West	with a common penalty	Giacomini-White	with a common penalty
0	18%	1.10	0.23	0.28	-2.22	-2.17
3	50%	1.60	7.09	7.06	3.18	3.16
5	44%	1.27	8.02	8.01	4.48	4.45
13	46%	1.11	8.09	8.04	5.34	5.31
21	33%	1.03	8.01	8.02	5.66	5.65
55	33%	0.95	9.21	9.20	7.02	7.03

The diagnostic runs from 0.95 to 1.60, and only $\delta = 3$ exceeds 1.5, the one Clark-West entry we would not lean on. The decisive column is the next one: imposing a common penalty moves no Clark-West statistic by more than 0.05, and changes no verdict at any delay. So the mechanism is present in principle but undetectable in practice here, and the column stands as a second opinion, reported with the ratio beside it instead of as a bare p-value.

The primary test is untouched, and for a reason, not by luck. The argument above turns on Clark-West being a test about *models*, so that anything changing the estimation penalty changes what its adjustment ought to be. Giacomini-White is a test about *methods* under a fixed rolling window, and a method is the whole procedure including whatever tuning happens inside that window. The validation slice is carved out of the training window, so it is part of what is being tested rather than a contaminant of a correction, and Giacomini-White makes no finite-sample correction that could be thrown off. The prediction is that its column barely moves under the remedy, and the last two columns of Table 7 are that prediction tested: no statistic moves by more than 0.06 and no verdict changes at any delay. The substitution rates, the bootstrap intervals and the simultaneous bands involve no nested-model correction at all and are unaffected.

A second concern is that we test ten delays at once. Ten pointwise 95% intervals will breach somewhere by chance, so we bootstrapped the whole vector jointly, in moving blocks of ten days to respect the overlap in the target [18], and took the largest studentised deviation across the grid. The resulting critical value is 6.51 instead of 1.96, a heavy penalty, and the effect still clears it from $\delta = 3$ to $\delta = 55$. One casualty is worth naming: the small negative reading at $\delta = 0$ does *not* survive the simultaneous band. We do not claim the cross-section is harmful when domestic data is current, and Section 5 gives the reason it is not: net of what it costs to estimate, its content there is zero.

The third concern is that the result is an artefact of fitting a wide model to a persistent series. We tested it by shifting the entire foreign block five years through the sample. The series keep their distributions, their autocorrelation, their crises; they lose only their alignment with the target. The substitution rate falls from +72% to -8%, and is negative at every delay we checked. Whatever the model is using, it is the alignment. The residual negative is not noise around zero and should not be read as one: it is the cost of breadth from Section 5, which is all that remains once the information is removed. A placebo of this kind lands below zero by construction, and a version of it that came back at exactly zero would be the surprising result.

Finally, Section 4 described the rate as slowing, which is a claim, not an observation. Comparing $R(55)$ with earlier values: against $R(13)$ the difference is +0.079 with an interval of [+0.037, +0.129], and against $R(21)$ it is +0.042 with [+0.014, +0.076], both still rising. Against $R(34)$ it is +0.018 with [-0.001, +0.039], which does not reject equality. The rate is still climbing out to about seven weeks and flat thereafter.

One last property of the rate was assumed and never tested. The rate is a ratio whose denominator is what the delay costs a domestic-only forecaster, and at short delays that quantity is near zero. A ratio with a denominator that can vanish is weakly identified, and a percentile bootstrap of it reports finite endpoints whether or not they mean anything. We therefore recomputed every interval by inverting a HAC t-test after Fieller, which returns two half-lines or the whole line when the data do not pin the ratio down. At every delay from one day to eleven weeks the confidence set is an ordinary interval, and Fieller's endpoints sit within 2.4 percentage points of the bootstrap's throughout: at $\delta = 55$, [63%, 79%] against [64%, 78%]. The objection is right in principle and inert here, so the published intervals stand. The Appendix gives the construction, the one delay where the ratio does not exist at all, and the error that revealed the first version to be wrong.

8. The race against implied volatility

The objection this section answers is the one Section 2 names as unavoidable: if a forecaster whose domestic data is δ days stale can see today's VIX, why would they need Tokyo? Implied volatility is forward-looking, it never goes stale, and both Korkusuz [9] and Buncic and Gisler [1] report that much of what looks like a cross-market effect is attributable to it. So the two candidates go into the same walk-forward, on the same labels, with the same estimator. Four models, all with the domestic block held δ days stale: domestic alone, plus the foreign block, plus implied volatility, and plus both. The number the objection turns on is what the foreign block adds *given* implied volatility. VIX enters at $t - 1$ throughout, the strictly admissible specification of Section 3; the same-day arm is reported below as a sensitivity and reaches the same verdict. VDAX-NEW begins in May 2001 against the index panel's February 2000, so all four are re-estimated on the same reduced window of 4,525 days and the comparison stays internal.

Table 8. Out-of-sample R^2 for four information sets, 4,525 days, September 2008 to September 2026. The last two columns are what the foreign block adds *given* implied volatility, and the Giacomini-White statistic testing it. VIX enters at $t - 1$, the strictly admissible specification of Section 3; the same-day timing sensitivity check follows in the text.

δ	domestic	+ foreign	+ implied vol	+ both	foreign IV	GW z
0	0.4958	0.4865	0.5379	0.5272	−0.0108	−2.48
3	0.3822	0.4270	0.5162	0.5053	−0.0109	−2.20
5	0.3350	0.4221	0.5151	0.5053	−0.0098	−1.92
13	0.1990	0.3928	0.5082	0.4989	−0.0093	−1.69
21	0.1082	0.3751	0.5068	0.4968	−0.0100	−1.81
55	−0.0369	0.3534	0.5088	0.4990	−0.0098	−1.73

The third column is the one that decides the section. A forecaster holding implied volatility loses almost nothing to delay: R^2 goes from 0.5379 to 0.5088 across eleven weeks, where the domestic-only forecaster falls from 0.4958 to −0.0369. Put the other way, a desk with VIX and VDAX and a three-week-old mark still beats a desk with fully current domestic data and no options information, 0.5068 against 0.4958. The last two columns are the comparison the objection turns on, and they are negative at every delay: adding seven foreign closes on top of implied volatility makes the forecast worse, not better, by about one point of R^2 . That reads like harm and Section 8.1 shows it is a bill, the cost of estimating seven coefficients, but on the question the section asks, the answer is already here. Breadth adds nothing measurable to a forecaster who can see the options market.

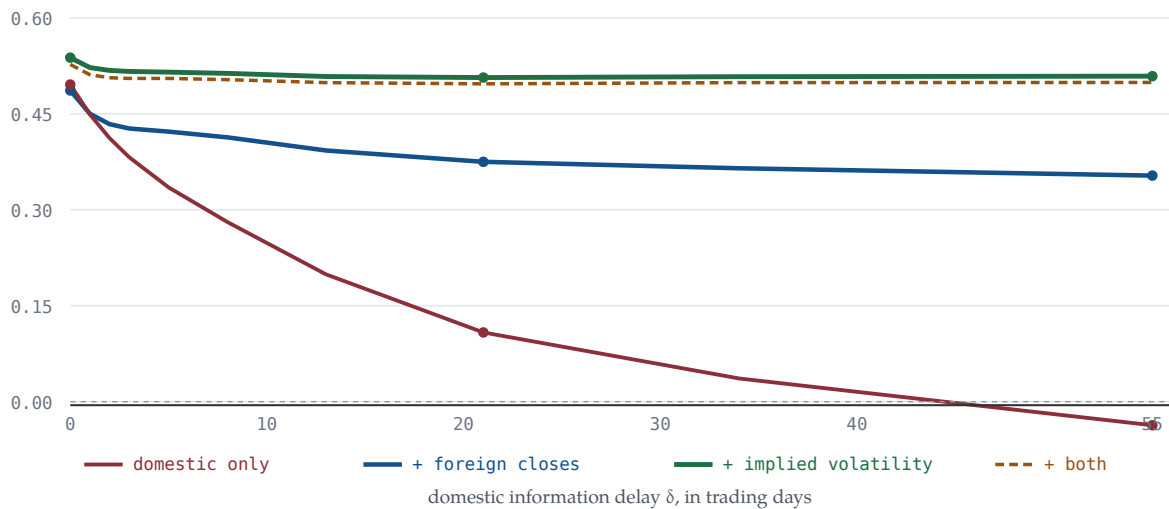


Figure 2. The green curve is almost flat. A forecaster holding implied volatility barely notices how old their domestic data is, and from about four days on they are ahead of where the domestic-only forecaster started. The dashed curve, which adds foreign closes on top, sits just below it at every delay.

The green curve is the result. Implied volatility does not merely slow the decay, it very nearly stops it: R^2 falls from 0.538 to 0.509 across eleven weeks, against 0.496 to −0.037 for the domestic model alone. Expressed as a substitution rate that is 103% at three weeks, and a rate above 100% is the signal from Section 3 that the measure has changed meaning. A forecaster holding three-week-old domestic data and today's VIX scores 0.507, above the 0.496 of a forecaster with fully current domestic data and no

options market. Implied volatility is not restoring what staleness took; it is supplying something the realised-variance history never contained.

That should not be mysterious, though it should not be overstated either. The VIX is a RISK-NEUTRAL expectation of S&P variance over roughly the coming MONTH; our target is REALISED variance over the coming five days. The two differ in horizon, by a factor of about four, and in measure, by a variance risk premium that is itself time-varying. So the VIX is not the quantity we are predicting. It is, however, a purpose-built forward-looking forecast of something close to it, and comparing that with a cross-section of correlated state variables is comparing a forecast with a set of clues. Read that way the surprise is not that implied volatility wins but that seven foreign equity markets get within two-thirds of it despite the horizon and measure mismatch working against the options market.

The sixth column is what the objection actually turns on, and it is negative at every delay. Given implied volatility, adding the foreign block makes the forecast slightly worse, and the Giacomini-White statistic, for which the domestic-plus-IV model is nested inside the full one so the same test applies as in Section 7, reaches conventional significance at the short delays. Taken literally this says foreign closes damage a forecast that already has VIX, which would be strange behaviour for genuine information.

Section 5 supplies the alternative reading, and the same instrument settles it. Replacing the seven foreign series with AR(1) surrogates matched on persistence and variance leaves regressors that carry nothing by construction, and what remains is the bill for estimating them. It comes to roughly 0.011 to 0.015 of R^2 , the whole of the negative. The content of the foreign block given implied volatility, gross of that cost, is +0.001 at $\delta = 0$, +0.007 at $\delta = 5$, +0.008 at $\delta = 21$ and +0.007 at $\delta = 55$, and bootstrapping each of those the way Section 5 describes gives intervals that all cover zero: $[-0.007, +0.010]$, $[-0.002, +0.017]$, $[-0.002, +0.019]$ and $[-0.003, +0.017]$. The correct word is redundant. Foreign closes do not damage a forecast that already sees the options market; they add nothing to it, and paying to estimate nothing is what makes the measured figure negative.

The table above is the admissible specification: VIX lagged a full day, which discards a day of information the live index would have carried. Substituting the same-day close, which contains fifteen minutes we are not entitled to, moves the options market the other way: R^2 at $\delta = 21$ rises from 0.507 to 0.568, a substitution rate of 119%, while the incremental contribution of the foreign block stays negative at all ten delays and its gross content stays indistinguishable from zero. Neither arm is the specification we would choose, and the one we would choose is not available to us. What licenses a conclusion here is not that the 21:00 observation falls between these two in time, which would say nothing about how it forecasts, but that the verdict is the same under a specification that gives the options market a full day less information and under one that gives it fifteen minutes more. A finding that survives that much variation in the input is not resting on the timing choice.

One feature of that result deserves a sentence, because it is more striking than it first looks. VIX is a thirty-calendar-day implied volatility and the target here is realised variance over five trading days, so the winning instrument is measured over a horizon six times longer than the thing it is forecasting. That it dominates anyway says something about how much of the five-day question is answered by the level of the whole term structure, not by its front. The matched instrument exists, since Cboe publishes a nine-day index, VIX9D, formerly VXST, and Section 8.5 runs the race again with it. The horizon does matter, by rather more than we expected, and the conclusion of this section does not.

8.1 Redundancy given VIX and VDAX, with one regressor instead of seven

The test above adds seven regressors to the implied-volatility model, and a referee points out that it therefore cannot separate two different reasons for a null: the foreign block carries nothing beyond VIX and VDAX, or it carries something and the cost of estimating seven coefficients consumes it. Section 5 says that cost is real and Section 5.1 says a compression avoids part of it, so the confound is not hypothetical. The compression supplies the fix: add *one* regressor instead of seven.

Table 9. What a single compressed summary of the foreign block adds on top of stale domestic data and the implied-volatility block, which is VIX and VDAX-NEW. The mean requires no estimation at all; PC1 is the real-time first component of Section 5.1. The seven-regressor column is Table 7's, for comparison.

δ	+ IV only	+ mean	+ PC1	+ all seven	mean IV	GW z	PC1 IV	GW z
0	0.5379	0.5392	0.5391	0.5272	+0.0012	0.50	+0.0012	0.48
3	0.5162	0.5184	0.5187	0.5053	+0.0022	0.98	+0.0025	1.07
5	0.5151	0.5179	0.5183	0.5053	+0.0028	1.02	+0.0032	1.12
13	0.5082	0.5120	0.5124	0.4989	+0.0037	1.16	+0.0042	1.26
21	0.5068	0.5106	0.5110	0.4968	+0.0038	1.19	+0.0042	1.28
55	0.5088	0.5129	0.5130	0.4990	+0.0041	1.31	+0.0043	1.30

The sign flips and the conclusion does not. With seven regressors the increment is negative at every delay; with one it is positive at every delay. That is the estimation-cost mechanism of Section 5 showing up exactly where it was predicted to, and it means the negative in Table 7 was a bill rather than evidence of harm, a reading the paper offered there and can now demonstrate rather than argue. But no Giacomini-White statistic reaches 1.96: the largest is 1.31, at eleven weeks. A single cheap summary of seven foreign markets, carrying essentially no estimation cost, still adds nothing detectable to a forecaster holding implied volatility.

That is a stronger statement than the one this section made before, and it should be read as strictly stronger, a sharpening of the earlier finding. The earlier test could be dismissed as penalising the foreign block for its parameter count. This one cannot. What it cannot do either is prove absence: increments of +0.001 to +0.004 with statistics between 0.48 and 1.31 are consistent with a small positive contribution this sample is too small to resolve, and the honest summary is that the foreign block adds nothing we can detect given implied volatility, not that it adds nothing.

8.2 Is redundancy an artefact of the linear estimator?

Redundancy was established with a ridge, and a referee may reasonably ask whether a linear map is simply too blunt to find what the foreign block holds given implied volatility. The test is to give the block a richer map and see whether anything appears: a random Fourier feature expansion, which is a one-hidden-layer network whose first layer is drawn, not learned, at two widths, and a k-nearest-neighbour arm that imposes no functional form at all.

Table 10. What each non-linear map of the foreign block adds to a forecaster holding stale domestic data and implied volatility. Wider penalty grid; Giacomini-White statistics, the implied-volatility model being nested in each. A statistic above +1.96 would overturn the redundancy result.

δ	R ² IV	+ foreign	GW z	+ FACTOR	GW z	+ RFF(20)	GW z	+ RFF(60)	GW z
0	0.5339	−0.0097	−2.47	−0.0111	−2.70	−0.0414	−2.95	−0.0538	−3.78
3	0.5066	−0.0051	−1.02	−0.0097	−1.84	−0.0457	−2.27	−0.0620	−3.19
5	0.5051	−0.0057	−1.04	−0.0076	−1.27	−0.0548	−2.48	−0.0709	−3.41
13	0.4955	−0.0054	−0.82	−0.0099	−1.02	−0.0652	−2.96	−0.0767	−3.49
21	0.4944	−0.0048	−0.79	−0.0161	−1.46	−0.0704	−2.83	−0.0830	−3.60
55	0.4903	−0.0048	−0.79	−0.0047	−0.70	−0.0614	−2.62	−0.0685	−2.91

No arm adds anything at any delay, and the two random-feature arms are significantly *worse* at all six. The direction of the capacity comparison is the informative part: RFF(60) is below RFF(20) at every delay, by 0.007 to 0.018 of R². If the objection were right, if there really were cross-market structure a linear map could not reach, more flexibility would find more of it. More flexibility finds less, and that is what over-parameterisation looks like when the signal is not there. The surrogate measurement puts a number on the mechanism: bending the foreign block costs 0.005 to 0.008 of R² for the FACTOR arm and 0.017 to 0.056 for the random-feature arms, rising with both delay and width.

One thing genuinely changes under the wider grid, and it favours the objection. The *linear* increment, significantly negative at short delays in Table 8, stops being significant from $\delta = 3$ onward, with statistics between −0.79 and −1.04. That is the right result and the one Section 5's estimation-cost argument predicted: the foreign block given implied volatility is not harmful, it is empty, and the apparent harm in Table 8 was the bill for estimating regressors that carry nothing. The redundancy conclusion is unchanged; its sign is better explained.

What this does not license is the claim that no non-linear model could find anything. The random first layer is not learned, and a learned one has strictly more free parameters, so it faces the same estimation-cost bill this sample has repeatedly failed to pay, which is an argument by extension rather than a test. Section 11 records it as such.

8.3 Is five days the result, or the setting?

Every number above forecasts five trading days ahead, and a reader is entitled to ask whether that is a finding or a convention. So the whole experiment is re-run at one day, two weeks and one month. One detail has to be settled before the table can be read: the Yang-Zhang proxy spans five days, which is how the latent variance is measured and has nothing to do with how far ahead we forecast, but it means that at $h = 1$ the outcome window shares four of its five days with the domestic feature the model already holds. Nothing leaks, since the outcome is still strictly in the future and every feature is still dated $t - \delta$ or earlier for the domestic block. But a forecaster at one day is being asked to predict an object they can largely already see, and their R² at $\delta = 0$ duly comes out at 0.921 against 0.499 at five days. The serial correlation of the target confirms it: 0.953 at one day against 0.692 at five, 0.576 at ten and 0.417 at twenty-one.

Table 11. The substitution rate at eleven weeks under four forecast horizons. Intervals are the full-ratio bootstrap of Section 4, in moving blocks of twice the horizon. The $h = 5$ row is the paper's headline and reproduces it exactly.

horizon	own R^2 at $\delta = 0$	$R(55)$ under R^2	95% CI	$R(55)$ under QLIKE
1 day	0.921	70%	[67%, 72%]	83%
5 days	0.499	72%	[64%, 79%]	81%
10 days	0.338	66%	[52%, 78%]	73%
21 days	0.161	51%	[23%, 71%]	65%

From one day to two weeks the rate is flat at 70%, 72% and 66% across a tenfold change in horizon and despite the one-day column measuring something partly mechanical. At one month it falls to 51%, the only point estimate outside the interval the headline quotes, though the intervals still overlap heavily and the one-month interval is wide enough on its own to be uninformative. QLIKE tells the same story more smoothly: 83%, 81%, 73%, 65%. So the headline is a property of the information over horizons out to a fortnight and begins to decay beyond that, which is what should happen, since a foreign close carries information about the next week's volatility state more readily than about the next month's, and by twenty-one days the domestic model it is being compared against has itself fallen to an R^2 of 0.161, so there is less loss left to recover and a noisier denominator to recover it against. We report the five-day figure as the headline because it is the horizon the whole paper is built on, not because it is the most favourable: at one day under QLIKE the rate is higher.

8.4 What the R^2 column means to someone sizing a position

A point of R^2 on log variance is not an intuitive quantity, and the reading a risk manager needs is narrower than the column suggests. What breadth repairs is a stale forecaster's tendency to over-estimate variance; the understating tail is unmoved: it runs between 8.6% and 11.4% of days whatever the delay and whichever model is used. Concretely, at eleven weeks the stale domestic forecaster's variance forecast is off by a median factor of 1.90, and is more than a factor of two out on 59.2% of days against 37.1% for the forecaster holding the cross-section, against 31.1% for a forecaster with no delay at all. Breadth closes most of that gap, and all of what it closes is over-estimation. Anyone reaching for the cross-section as protection against being surprised by a volatility spike is reaching for the wrong instrument.

8.5 The same race, with the horizons matched

The race above has an asymmetry in it. The target is realised variance five trading days ahead; VIX prices a thirty-day horizon. A thirty-day option-implied variance is not the natural forecast of a five-day realised variance, so the section was run with a horse suited to a different distance. Cboe publishes VIX9D, a nine-day index on the same underlying computed the same way, from January 2011. Nine days against five is not exact either, but it is the closest listed instrument that exists and it is four times closer than thirty. Both blocks are given the same number of regressors, one implied-volatility series each plus VDAX-NEW on both sides, so the estimation cost of Section 5.2 is identical on the two sides and cancels in the difference.

The cost of the test belongs before its result. VIX9D begins in 2011, the decision variable of equation (2) needs a 252-day trailing median before it exists, and the walk-forward needs 1,500 days of labelled history before it can fit, which leaves 2,098 test days from May 2018, 46% of Table 8's window. It contains the 2020 episode but not the 2008 one, and 2008 is where being late costs a forecaster most.

That sample is also simply harder: the identical thirty-day block loses between 0.0151 and 0.0233 of R^2 when it is moved to the shorter window with nothing else changed. We measure that separately so it cannot be mistaken for the horizon effect, and every column below sits inside the same window.

Table 12. The horse race with the horizons matched, 2,098 days, May 2018 to September 2026. IV30 is the block of Table 8, VIX at $t - 1$ with VDAX-NEW; IV9 replaces VIX with VIX9D at the same regressor count; TERM carries both tenors. The fifth column is the horizon effect on a fixed sample and the last is what the seven foreign closes add *given* the horizon-matched block. Giacomini-White throughout: IV9 against IV30 is a comparison of two non-nested models of equal width, and the foreign increment is nested.

δ	domestic	+ IV30	+ IV9	+ TERM	IV9 – IV30	GW z	foreign IV9	GW z
0	0.4720	0.5165	0.5408	0.5376	+0.0243	4.06	−0.0141	−2.41
5	0.2979	0.4918	0.5321	0.5288	+0.0403	4.19	−0.0179	−3.34
13	0.1696	0.4893	0.5303	0.5246	+0.0410	4.13	−0.0162	−3.38
21	0.0921	0.4906	0.5304	0.5244	+0.0397	4.02	−0.0155	−3.17
55	−0.0600	0.4936	0.5332	0.5312	+0.0396	4.06	−0.0138	−2.59

Two things are in that table and they point in opposite directions. The horizon objection is correct: matching it is worth between 0.0243 and 0.0410 of R^2 , at every one of the five delays, with Giacomini-White statistics of 4.02 to 4.19. This is not a marginal improvement that survives on a technicality; the nine-day index is simply the better instrument for a five-day target, and Section 8 had been using the worse one. But the conclusion the section draws does not move. The last two columns are what the foreign cross-section adds on top of the horizon-matched block, and they are negative at every delay, with statistics between −2.41 and −3.38. The paper had been reporting redundancy against an implied-volatility control that was weaker than it should have been, and redundancy holds against the stronger one. A referee who makes the horizon objection is right about the instrument and gets a firmer version of the same answer.

Two details close off the obvious follow-ups. Replacing the nine-day level with the nine-to-thirty-day slope, at the same regressor count, is worse at every delay, with statistics from −2.29 to −4.15: what the near end of the term structure carries is its level, not the shape of the curve. And carrying both tenors is slightly worse than carrying the nine-day one alone, 0.5376 against 0.5408 at zero delay, because the second series does not pay for its own coefficient, which is Section 5.2's mechanism again on a two-column block. One incidental is worth recording because it was not designed: on this window, which shares barely half its days with Table 1's, the foreign block's substitution rate comes out 51%, 64%, 68% and 73% at five, thirteen, twenty-one and fifty-five days, against 51%, 64%, 67% and 72% in Table 1. $R(\delta)$ barely moves between two half-overlapping samples measured eight years apart.

One objection to all of this has to be answered rather than noted, and it is the obvious one. This window contains exactly one violent episode, and a front-tenor index separating from a back one is precisely what a crisis in the term structure would produce, so the whole of Table 12 could be a fact about March 2020. The test is to delete it: every test day from 15 February to 30 June 2020 is removed from the evaluation, the models untouched, leaving 2,004 days. The nine-day index still wins at all five delays, by 0.0264 to 0.0432 of R^2 with statistics of 3.37 to 3.64, larger at four of the five delays than with 2020 included rather than smaller, and the foreign block still adds nothing, between −0.0158 and −0.0212 with statistics from −2.11 to −3.26. Neither conclusion in this subsection rests on the crisis.

9. Where each result holds

The same exercise with each of five indices as the target is in Table 19, on the continuous target and at every delay.

The effect appears everywhere, and the Nikkei is the case that carries information. It closes before every other market in the sample, so it has no same-day foreign observation at all, since every peer reaches it a session late, and it recovers the least, 60% against 69% to 88% elsewhere. That is what the mechanism predicts and what a spurious correlation would not produce. Past that contrast we can say little: the S&P has six same-day peers and recovers less than the FTSE with four, the intervals overlap heavily, and five markets cannot resolve an ordering.

One further objection deserves an answer. Rapach, Strauss and Zhou find that lagged US returns predict foreign returns while the reverse is weak [16]. That concerns returns while ours concerns volatility, and lagged returns where ours are same-day closes observed before the target's own close. We rely on both distinctions, and a reader who rejects them should discount Sections 4 through 6 accordingly.

The scope condition matters more than the ordering. Section 8 shows that where a liquid options market exists on the asset, breadth is redundant, and the S&P 500 is the most heavily optioned index in the world. So the breadth result may be most relevant to assets without such a market, the case Section 1 opens with: a private portfolio marked quarterly, a property book on an appraisal cycle, a fund reporting NAV with a lag. Nobody writes a liquid option on those. What a holder of them can see is the public cross-section, and Sections 4 through 6 say what that is worth as a function of how stale the mark has become.

We would rather state that plainly than dress it up, because it is also the weakest point of the design. We ran the experiment on a liquid asset in order to observe the truth, and the property that makes the truth observable, a deep and continuously traded market, is the same property that brings an options market with it. The result we can measure cleanly and the case we care about are therefore separated by the one feature that decides which substitute wins. Section 11 says what would close that gap.

There remains the $\delta = 0$ row of Table 1, and it is the one place in this paper where the choice of loss decides the sign. On log-scale squared error the cross-section is worse by 0.0103, interval $[-0.0192, -0.0016]$; on QLIKE it is worse by 0.0142, $[-0.0292, -0.0023]$; on squared error on the variance scale it is BETTER by 0.0071, $[-0.0470, +0.0280]$. Two of the three exclude zero and point one way, the third does not exclude zero and points the other. An earlier draft read the negative as settled and used it to press a disagreement with Korkusuz [9] and Jayawardena et al. [6], who find foreign volatility improves a current forecast. We withdraw that. A sign that moves with a monotone transformation of the target is not a finding about the world, and the two losses that agree are the log-scale one, which is not proxy-robust, and QLIKE, which shares the log-ratio structure. The honest statement is that at zero delay our design cannot tell which way the effect runs, and nothing here bears on their result. Table 2's decomposition still holds conditional on the log scale: whatever is there is small, bounded near one point of R^2 , and mostly the bill for estimating seven coefficients.

9.1 In stress and in calm

Everything above is an average over 4,862 days, and a referee objects that the interesting days are not average: a rolling median assumes a stationary volatility distribution, and across a structural macro transition it should fail to adjust. The objection is testable here because the test window contains the 2008 crisis, 2011, August 2015, February 2018, COVID and the 2022 inflation shock. It is not testable on the companion note, whose window begins in November 2023 and contains no transition at all, and the note says so instead of answering it.

Two decisions shape what follows. The first is that a single split is not enough. Six named episodes at full width run to 866 days, most of them ordinary: "the 2022 inflation shock" as the whole of 2022 is 251 days. A wide window dilutes exactly what is being tested. So the definition is tightened in rungs: VIX at $t-1$ above its own trailing 67th, 90th and 95th percentile, which needs no dates and is available to a forecaster in real time, together with the named episodes at full width and as the 41 trading days centred on each window's VIX peak. The tightening was added after the wide version had been run, and we say so instead of presenting it as the original design.

The second decision matters more. Squared errors are larger when the target is more volatile, so a larger loss differential in stress can be nothing but heteroskedasticity. Every figure below is therefore the differential divided by the domestic model's own mean squared error *inside the same regime*: a share of error removed, comparable across states in a way the raw difference is not. The raw column is reported beside it, and the two tell different stories, and that is the reason for the precaution.

Table 13. What the cross-section removes from the domestic model's error, by market state, at $\delta = 55$. Rungs are nested: the top 5% of VIX days is a subset of the top 10%. Intervals are moving-block bootstraps of the ratio, taken on each regime's own time-ordered subsequence.

state	test days	share of domestic MSE removed	95% interval
VIX top 5%	367	63.1%	[+51.3%, +72.6%]
VIX top 10%	644	57.0%	[+46.1%, +66.3%]
VIX top 33%	1,729	43.3%	[+32.9%, +52.5%]
episodes, peak ± 20 days	246	63.1%	[+45.6%, +75.7%]
VIX bottom 67%	3,074	29.9%	[+23.8%, +35.5%]
outside every episode	3,996	23.9%	[+17.0%, +29.8%]

The objection is answered in the opposite direction. The contribution does not weaken in stress; it roughly doubles, and it rises monotonically as the definition tightens, running 43.3%, 57.0% and 63.1% along the VIX ladder against 29.9% on calm days. The severest rung and the calmest have disjoint intervals. The calendar route, built from dates instead of percentiles, lands on the same 63.1% at its narrow window, which is a useful agreement because the two cuts share no construction. A dose-response along a mechanical ladder is stronger evidence than any single split would have been, and we would not have had it from the version of this test we ran first.

The economic reading is the one a risk manager would want, and it is not the reading Section 8.4 gave. Breadth is worth most precisely when the domestic mark is stale *and* the market is moving, which is when a stale mark is most dangerous. That is a stronger claim than the paper made before, and it belongs to the cross-section itself and to no refinement of it.

9.2 Does implied volatility stop dominating when the options market is stressed?

The mirror objection to Section 8 is that redundancy given VIX and VDAX assumes a well-functioning options surface, and that during a crunch breadth might stop being redundant exactly when it matters. Applying the same ladder to the increment the foreign block adds on top of the implied-volatility model, across four delays and six states, the foreign block adds significantly in none of twenty-four cells, on the stressed rungs and the calm ones alike, down to the top 5% of VIX days. Redundancy given implied volatility is not an average concealing a stressed subsample. The severest rung carries 324 days, so an effect confined to conditions more extreme than the ladder reaches would not appear here.

9.3 What if the foreign feed is late too?

Every number to this point was computed with the foreign block at its freshest admissible value: for each peer, the most recent close that precedes the target's. That is a deliberate choice about admissibility and it has been doing more work than the paper admits. A desk holding a stale mark is, almost by definition, a desk with a data problem, and the infrastructure that delays a domestic mark by δ days is not obviously the kind that delivers seven foreign closes the same evening. Nothing above says what such a desk gets. A recommendation that assumes a same-day feed without saying so is worth less than it appears, so the assumption is removed here instead of defended.

Let g be the number of days by which the foreign feed is itself late, and repeat the walk-forward of Section 4 at every pair (δ, g) . The domestic block is held δ days stale as before; the peers are held g days stale on top of admissibility. The column $g = 0$ is the paper.

Table 14. Substitution rate $R(\delta)$ when the foreign block is itself g days stale. Same walk-forward, same losses and same test days as Table 1; $g = 0$ reproduces it. A negative rate means the model carrying the aged cross-section forecasts worse than the stale domestic block alone. Undefined at $\delta = 0$, where there is no delay-induced loss to recover.

δ	$g = 0$	$g = 1$	$g = 2$	$g = 3$	$g = 5$	$g = 10$
3	36%	9%	-8%	-15%	-14%	-10%
5	51%	29%	13%	-0%	-11%	-8%
13	64%	52%	42%	34%	22%	-1%
21	68%	57%	50%	44%	33%	16%
55	71%	64%	58%	53%	46%	30%

The first column is Table 1's and the rest are not near it. Ageing the foreign feed reduces skill significantly in 25 of the 30 cells tested, and in 5 of the 6 delays a single day is already enough. At $\delta = 5$ the rate falls from 51% to 29% with one day and to zero with three. At $\delta = 3$ the cross-section becomes actively harmful from $g = 2$ onwards, the honest description of a negative rate: the delayed desk would have done better ignoring it.

Shares of a loss are a poor way to deliver that, so here it is in the unit this paper uses everywhere else: the age of a domestic mark. The own-only delay curve is re-estimated on a fine grid, and each combined model is quoted as the age of the plain stale mark that forecasts exactly as well.

Table 15. Effective age. Each cell is the age of a plain domestic mark whose own-only forecast matches the model that has a δ -day-old mark plus a foreign block g days stale, by interpolation on the own-only delay curve over sixteen ages from 0 to 55 days.

δ	$g = 0$	$g = 1$	$g = 2$	$g = 3$	$g = 5$	$g = 10$
3	1.7d	2.7d	3.3d	3.7d	3.6d	3.4d
5	1.9d	3.0d	4.0d	5.0d	5.9d	5.6d
13	2.8d	4.2d	5.5d	6.8d	8.9d	13.2d
21	3.5d	5.2d	6.7d	8.1d	10.5d	15.0d
55	4.6d	6.8d	8.4d	10.0d	12.6d	19.4d

The last row states this paper's headline in one number: a fifty-five-day-old mark plus today's foreign closes forecasts like a mark 4.6 days old. The same mark with yesterday's foreign closes forecasts like one 6.8 days old, and with the feed a week behind, 12.6 days. Because this is the sentence the paper will be quoted on, it carries an interval like every other headline here, and constructing one takes some care: the map from skill to age is the inverse of a delay curve that is steep near zero and flat far out, so a fixed error in R^2 becomes a small error in age at one end and a large one at the other, and the interval cannot be obtained by pushing an R^2 interval through the curve. Resampling test days in moving blocks and re-interpolating inside each replication, recomputing the curve, the combined model and the benchmark variance on the same resampled days, gives 4.6 days [2.8, 7.7]. The one-day and week-behind feeds give 6.8 [4.4, 10.2] and 12.6 [9.7, 17.5]. No replication placed the model at a fresh mark or off the end of the grid, so the interpolation is not being asked to extrapolate.

The same construction prices the feed itself. Differencing the ages *within* each replication, a single day of foreign staleness costs 2.1 days of domestic freshness at eleven weeks, interval [1.4, 3.3], and five days cost 8.0 [5.5, 12.0]. Both exclude zero. The marginal intervals in the previous paragraph overlap, as marginal intervals ordinarily do and is not evidence against a paired difference; the paired construction is the one that answers the question, and it agrees with the paired skill test reported above. Fitting a line along each row of Table 15 gives the same exchange rate averaged over the whole row instead of at its first step. It runs 0.13, 0.35, 1.03, 1.13 and 1.44 days at $\delta = 3, 5, 13, 21$ and 55, rising at every step. The shape is the point. Where the mark is nearly fresh the cross-section is decoration and ageing it costs almost nothing; where the mark is old the cross-section is doing the work, and then a day of the foreign feed costs about a day of the domestic one. The readers with most to gain from Sections 4 to 6 are exactly the ones whose foreign data must be current.

One reading of that decay has to be ruled out before the rest is safe. If the foreign block worked by propagation, with volatility travelling from Tokyo to New York overnight, its value would be its earliness, the decay above would be the wave passing, and the right response would be a richer dynamic model, not a warning about feeds. The cross-correlations appear to speak to this and they are a trap. Correlating each peer at $t-1$ against the target at t , then the target at $t-1$ against the peer at t , all seven peers appear to LAG the target. But a row of this panel is not an instant. The target closes at 21:00 UTC and a peer closes c hours earlier, so the first direction spans $24 + c$ hours and the second $24 - c$; for a persistent series the shorter span gives the higher correlation, and the clock alone predicts the observed sign whether or not anything leads anything.

That prediction is quantitative, so it can be tested instead of conceded. Each peer's asymmetry should scale with $2c$. Across the seven it does, with a correlation of +0.93: 0.013 and 0.019 for the FTSE and

the DAX, four and a half hours ahead of New York, against 0.051 for Bovespa, a full day ahead once admissibility has shifted it. The apparent lead-lag structure is a property of the calendar, not of the markets, and it is evidence for neither story. A check from the other side agrees: replacing the seven peers with the single real-time global factor of Section 5.1 and entering it at one lag at a time, the contemporaneous factor beats every lagged version in 26 of 30 cells. Yesterday's factor does add to today's at $\delta = 3$ and $\delta = 5$, and at no longer delay, so a little genuine dynamics is present, inside an otherwise contemporaneous story.

What is left is the reading Section 5.3 already arrived at by a different route. The block is a second reading of one shared state, it loses value as it ages at about the rate the first reading does, and it stops contributing once it is older than the mark it was brought in to replace: at $\delta = 13$ a foreign block ten days stale returns the model to an effective age of 13.2 days, the domestic mark on its own. Breadth substitutes for a stale mark by being fresh, not by being early.

10. What the rate depends on

$R(\delta)$ carries S_{own} in three of its four places, so every figure above is a statement about the domestic model as much as about the foreign one. That is uncomfortable, because the paper has spent its effort on the other side: Section 5.1 compresses the foreign block, Section 5.3 splits it, Section 8.1 tries it as one regressor. The control has been the same three features throughout. This section varies it, and then varies three other things that could make the headline wrong: the sample period, the number of models we tried before arriving at this one, and the gap between training and forecast. Each was run once and is reported whatever it said.

10.1 The control, and what the normalisation hides

Five stronger domestic blocks were built, each a superset of the paper's: four further own lags; quarterly and annual means; leverage terms from the signed return; a second realised measure together with the LEVEL of the trailing median; and all of them at once. Each was given the same walk-forward, the same refit schedule and the same ridge, and $R(\delta)$ was recomputed inside each arm, so every rate is measured against its own control. The strongest was chosen on own-only skill averaged across delays, never on the substitution rate, the quantity under test and must not be allowed to select its own control.

Table 16. The paper's domestic block against the strongest alternative. The first two columns are out-of-sample R^2 for the domestic model alone, δ days stale. The rate is recomputed inside each arm. The interval is on the DIFFERENCE, from a paired bootstrap that recomputes both ratios, numerator and denominator, on the same resampled days.

δ	R^2 own, paper	R^2 own, +level	$R(\delta)$ paper	$R(\delta)$ +level	difference	95% interval
0	0.4987	0.5050	n/a	n/a	n/a	n/a
5	0.3371	0.3505	51%	47%	-4.0%	[-10.0%, +0.8%]
13	0.2011	0.2226	64%	60%	-3.5%	[-7.3%, -0.2%]
21	0.1121	0.1448	67%	64%	-3.3%	[-6.8%, -0.5%]
55	-0.0351	0.0172	72%	68%	-4.1%	[-7.2%, -1.4%]

The control was improvable and the rate moves. A richer domestic block beats the paper's in 14 of 45 cells, and the substitution rate falls by three to four points, significantly at $\delta = 13, 21, 34$ and 55. Four

points is not a reversal, since 68% sits inside the [64, 79] interval Table 1 already carries, but the paired test is sharper than the overlap and it excludes zero, so the gap is real and the paper quotes both figures and leans on neither.

What makes that arm strong is the part worth keeping. It carries two features and they are not the same kind of thing. One is a second realised measure, a 22-day Yang-Zhang, which is simply less noisy than the 5-day one; the other is $\log M_t$, the LEVEL of the trailing median. Splitting them settles it at once. At eleven weeks the paper's block scores -0.0351 , adding the second measure scores -0.0428 , which is worse, and adding the level alone scores $+0.0308$. The level does the work, at nine delays out of nine, and by a margin that grows with staleness: $+0.0011$ of R^2 at zero delay against $+0.0736$ at eleven weeks.

That is a finding about this paper's own design, not about volatility. Equation (2) divides every series by its own trailing median, and that is what puts Tokyo, São Paulo and the VIX on one scale and makes a cross-section possible at all. It is also what prevents every model in the paper from seeing how volatile the world actually is. A forecaster who has lost his recent observations still knows roughly what regime he was in, and that knowledge is worth more the more he has lost, which is where the paper claims its largest effect. The comparability the design buys is paid for in domestic information, the price rises with delay, and we have not found this trade priced anywhere in the literature the paper cites. Anyone normalising a panel this way should measure what it costs before assuming it is free.

That claim has an obvious objection and it has to be answered rather than noted. The target is $\log(\sigma^2 / M_{t+5})$, M is a 252-day median and therefore very slow, so a model handed $\log M_{t-\delta}$ could lower its squared error simply by predicting the target's own DENOMINATOR. That would be arithmetic about how the target was built, not a forecast of volatility. Nothing would be inadmissible, since every feature is dated $t - \delta$, but the finding would be empty.

The test is to re-date the normaliser and see what survives. Three targets were built, differing in nothing else: the paper's, normalised at $t+5$; the origin-dated one of Section 3, normalised at t ; and one normalised at $t - \delta$, the forecaster's own information set. The third is decisive, because its denominator is known exactly when the forecast is made, so a model cannot buy anything by predicting a quantity it has already been given, and it is also the only one of the three a delayed forecaster could actually use. Across the four positive delays the third target retains between 87% and 102% of the gain: $+0.0680$ against $+0.0667$ at eleven weeks. Re-dating the normaliser changes nothing, so the level is not being used to undo the normalisation. It is information.

Two qualifications. The gain reaches conventional significance only at $\delta = 55$, on all three targets alike, so the short-delay column shows a direction without establishing a result. And the rate correction above survives the same test: on the implementable target the level moves $R(55)$ from 74% to 70%, a shift of the same size as the 72% to 68% reported here.

10.2 Is the rate stable over nineteen years?

Section 9.1 conditions on market state; this asks the different question of whether the result drifts in time. Split at the midpoint, the two halves agree at every delay, 77% against 67% at eleven weeks, with overlapping intervals at all four. A rolling window tells a less settled story: $R(5)$ runs from 22% in the window ending April 2019 to 91% in the window ending April 2015, and a line through those

points falls about 22 points a decade at five days and 12 at eleven weeks. Those windows overlap by 750 of 1,000 days, so the path is descriptive and no standard error is quoted for it.

The obvious explanation is composition, since the first half holds 2008, 2011 and 2015 and Section 9.1 has already shown the rate is higher in stress. Holding stress fixed settles it: comparing the halves within a VIX tercile, not one of twelve cells differs significantly. The drift in the path cannot be attributed to a change in how the cross-section works, and on this sample it cannot be attributed to anything.

Conditioning does something else, though, and it is the reason Section 4 quotes the rate by state.

Table 17. The substitution rate by market state, pooled over the whole window, VIX terciles taken at $t-1$. Same whole-ratio block bootstrap as Table 1. A calm-market rate is imprecise by construction: there is little delay damage in a calm market, so the ratio's denominator is near zero.

δ	calm	middle	stressed
5	42% [-4, 102]	32% [8, 51]	64% [49, 81]
13	43% [-6, 100]	47% [29, 61]	74% [64, 84]
21	41% [-10, 83]	53% [36, 66]	77% [68, 85]
55	47% [6, 84]	53% [37, 67]	81% [74, 87]

The stressed column is the one this paper can actually report: 81% at eleven weeks with an interval of [74, 87], and 81% in each half of the sample taken separately. Before leaning on that, one objection has to be cleared. $R(\delta)$ divides by the damage the delay does, and stressed days are by definition the days on which being late costs most, so stress is where the denominator is largest, and therefore where a ratio with that denominator underneath is best behaved. A high, tight rate in stress could mean the cross-section does more work there, or merely that the ratio is measurable there. The first is a claim about markets; the second is true of any ratio and says nothing.

Taking the ratio apart settles it. Write $N(\delta)$ for the mean squared error the cross-section removes and $D(\delta)$ for the mean squared error the delay adds, both on the same days and in the target's own units, so that $R = N / D$.

Table 18. The substitution rate taken apart, by market state. N is the error the cross-section removes, D the error the delay adds, both mean squared error on the log-variance target. If only D moved with state, the conditional result would be arithmetic, not a finding.

δ	N calm	N stressed	N ratio	D calm	D stressed	D ratio
5	0.0284	0.1567	5.5×	0.0673	0.2434	3.6×
13	0.0345	0.3787	11.0×	0.0798	0.5110	6.4×
21	0.0412	0.5224	12.7×	0.1003	0.6755	6.7×
55	0.0583	0.8043	13.8×	0.1234	0.9927	8.0×

Both move, and the numerator moves further: averaged over delays, stress multiplies N by 10.7 and D by 6.2. The cross-section removes more error in absolute terms when markets are frightened, not merely a larger share of a larger loss, so the conditional result is a fact about markets. At eleven weeks it removes 0.8043 of squared error in stressed markets against 0.0583 in calm ones.

So the conditional reading stands, and on absolute error as well as on the ratio. The calm column is not a small number, it is an imprecise one: it spans most of the unit interval and at three of the four delays it does not exclude zero. We suspected worse. A ratio whose denominator can approach zero need not have a bounded confidence set at all, and a percentile bootstrap cannot express that, so the Appendix's Fieller construction was applied to all twelve cells. Every one comes back an ordinary interval, and within a point or two of the bootstrap's: the calm cell at five days is $[-4\%, +104\%]$ against the bootstrap's $[-4\%, 102\%]$, the stressed cell at eleven weeks $[+74\%, +87\%]$ against $[74, 87]$. The denominator never approaches zero closely enough to misbehave, so the printed intervals stand and the calm rate is imprecise without being unidentified. That is not a failure of the estimator but a property of the question: where delay costs a forecaster almost nothing, the share of that cost which the cross-section repairs has no stable value. An unconditional rate averages a well-measured result with an unmeasurable one, and Section 4 says so instead of quoting the average alone.

Conditioning on the VIX raises a fair objection, since Section 8 says that a forecaster who holds the VIX should ignore the cross-section altogether. The two are compatible, but only for a specific reader, and it is worth being exact about who. The VIX is a public index on the S&P; a holder of an illiquid asset can see it whether or not any option trades on the thing they own. Such a reader can therefore condition on market state without being able to buy the substitute that state variable makes redundant. A reader who holds options on their own asset is in Section 8's case and should stop reading at Section 8. We also note that Section 9.1 works with a ladder of cuts at the 33rd, 90th and 95th percentiles while this section uses terciles, and that the tercile split was chosen after that ladder had been seen.

10.3 How many models were tried

Thirty-three scripts stand behind this paper and nine distinct forecasters have been fitted to this target across them. A single test statistic on the one that won is not the right instrument for that, so the whole set is put through White's Reality Check and Hansen's SPA, with the size of both first validated by simulation on this sample's own dependence structure, where a nominal 5% test rejects 5.5% and 7.0% of the time respectively, with SPA rejecting more often, as Hansen predicts. Against the domestic benchmark, the best of the nine survives both tests at $\delta = 3, 5, 13, 21$ and 55 with p below 0.0005. It fails at $\delta = 0$, where the best margin across all nine models is +0.0014 of R^2 : nothing beats the benchmark there at all, so that failure is an absence of any effect to correct.

10.4 Leakage from an overlapping target

The target is realised variance over the next five days, so consecutive observations share four days of outcome, and a naive split would put part of the answer in the training set. Every refit here already cuts the training window at $t - \delta - h$, which removes exactly the rows whose outcome was unresolved at the forecast date: a purge, though the paper has not called it one. Removing it inflates R^2 by +0.0024 [+0.00062, +0.00387] at zero delay and +0.0021 [+0.00017, +0.00382] at three days, so it is load-bearing and any study of a five-day target that splits without one is reporting inflated numbers. Adding an embargo on top of it costs nothing detectable in any of eighteen cells, so the gap the design already leaves is sufficient.

10.5 Five markets, at every delay

The five-market comparison Section 9 points to is below, across the whole delay curve, not at one point, and it is where the timing mechanism shows most plainly.

Table 19. Substitution rate by target market at four delays, continuous target, R^2 scale. Same-day peers are the markets closing earlier on the same calendar day; every other peer contributes the previous session. Intervals bootstrap the whole ratio.

target	close, UTC	same-day peers	R(5)	R(13)	R(21)	R(55)
S&P 500	21:00	6	51% [37, 66]	64% [53, 74]	67% [58, 76]	72% [64, 79]
FTSE 100	16:30	4	74% [56, 90]	82% [70, 92]	84% [74, 93]	88% [80, 94]
DAX	16:30	4	55% [41, 68]	69% [58, 78]	71% [62, 78]	74% [67, 80]
Hang Seng	08:00	2	61% [34, 86]	68% [50, 86]	69% [54, 84]	69% [58, 80]
Nikkei 225	06:00	0	44% [17, 71]	54% [33, 73]	55% [36, 73]	60% [45, 73]

The ordering by same-day peers is not perfect, since the FTSE has four and recovers more than the S&P with six, but the two ends are where the mechanism predicts. The Nikkei closes before every other market in the sample, has no same-day foreign observation at all, and recovers least at every delay; the FTSE and DAX, which see all of Asia before they close, recover most. The curves are also flatter than the S&P's: for markets with early peers the rate is already near its eleven-week value within a week.

11. What we cannot say

The limitation that matters most is the one Section 9 names. The illiquid-asset case that motivates the whole exercise is argued rather than demonstrated, and Section 8 raises the cost of that gap rather than leaving it where it was: we now know the answer depends on whether an options market exists on the asset, and we have tested only the side where one does. Section 13 narrows the gap by one step, showing that the clean-lag model of staleness behaves like periodic marking at matched mean staleness, so the apparatus is at least the right shape for the case we cannot reach. It does not close it. Only a genuine appraisal-based or quarterly-marked target, on an asset with no options written on it, would do that.

Section 9.3 adds a limitation that applies to every number in this paper and to no single section of it. All of them were computed with the foreign block at its freshest admissible value, and the paper did not say so because it did not occur to us to vary it. A desk whose own foreign feed is one day behind recovers 29% of the delay-induced loss at five days where Table 1 reports 51%, and one three days behind recovers none of it. The measurement is not wrong, but the claim is narrower than it reads: it describes a forecaster with a stale domestic mark and a current foreign feed, a more particular creature than the one Section 1 has in mind. Whether an illiquid-asset holder is that creature is a question about their data vendor, not about volatility, and we cannot answer it.

One limitation from Section 10 qualifies the headline itself: the unconditional rate averages two states in which the mechanism behaves differently. In stressed markets it is well determined, stable across both halves of the sample, and large in absolute terms as well as in share. In calm markets it is imprecise enough to be uninformative. Table 1 is an average of the two, and the conditional figures in Table 17 are what a reader should carry.

The implied-volatility block is thinner than we would like. VIX and VDAX cover the sample. A spot VSTOXX index was obtained and tested, and it cannot join the headline block because requiring it costs 73% of the sample: insisting all three series be present moves the window from 4,525 days beginning September 2008 to 1,240 days beginning April 2020, which discards the 2008 crisis outright and begins after the March 2020 spike: peak VIX is 82.7 over the full sample and 41.4 inside what remains. Holding those 1,240 days fixed and varying only the block, the third series adds nothing at any delay and is mildly negative at all six, because VSTOXX and VDAX are both European equity volatility indices;⁴ reports what it shows; the short version is that it does not cover the sample either, so the headline block stays at two series. On the window where three series are available, the third one makes the options arm slightly worse, because VSTOXX and VDAX are both European volatility indices and the third column costs more to estimate than it returns. The conclusion it was defending, that the foreign block is redundant given implied volatility, is unaffected, so the error was worth correcting rather than quietly dropping.

The non-linearity result in Section 8.2 is bounded in a specific way. A random Fourier layer is a one-hidden-layer network whose first layer is drawn rather than learned, and no network with a learned first layer was fitted. The argument that one would not change the answer is an extension of the estimation-cost finding: more free parameters, the same bill, a sample that has repeatedly failed to pay it, and a capacity comparison already pointing the wrong way. An extension is not a test. Section 8.4 names a limitation of a different kind, and it is the one a practitioner should read first: what breadth repairs is a stale forecaster's tendency to over-estimate variance, not their exposure to under-estimating it. The understating tail is unmoved at every delay we checked. Anyone reaching for the cross-section as protection against being caught short by a volatility spike is reaching for the wrong thing. Section 5.3 renames the mechanism and inherits that renaming's own limit: a first principal component of seven equity indices is a proxy for global systemic risk, not a measurement of it, so orthogonalising the block against it is weaker than orthogonalising against a factor built from a wider universe, and the remainder may still carry global content this basis cannot see. The direction of that test is informative; its magnitude is not a bound. Section 9.2 has a limit of the same shape: its severest rung holds 324 days, so a contribution confined to conditions more extreme than the ladder reaches would not appear, and nothing there rules one out.

The largest of these limits is the estimator, and we can be specific about it rather than gesturing, because we fitted the alternative. Every model in this paper is a ridge on a fixed feature set, chosen because it is transparent and because the object under study is the substitution rate, not the best attainable forecast. The natural competitor is a state-space model: one latent global volatility factor on an AR(1), every series loading on it, and a Kalman filter handling a missing observation by dropping that row from the measurement equation, which is what a delayed domestic mark *is*, not something to be imposed on the data. Fitted two-step after Doz, Giannone and Reichlin [21], with the state-space parameters estimated once on the initial training block and then frozen, and the filtered state fed into the same forecasting ridge on the same days, it beats the specification used throughout this paper at $\delta = 21$ by 0.0124 of R^2 , interval $[+0.00114, +0.02192]$, and at $\delta = 55$ by 0.0204, interval $[+0.00728, +0.03109]$. It loses at $\delta = 0$, by 0.0233, interval $[-0.03280, -0.01047]$, and the crossover is the interesting part: imposing a one-factor structure costs skill when the mark is fresh and pays when it is stale. Its reconstruction of the current domestic state that the forecaster is not allowed to see correlates 0.84 with the truth at eleven weeks, against 0.18 for the stale value itself.

We report this rather than adopting it, and the reason is a scope judgement a reader is entitled to disagree with. $R(\delta)$ is a ratio in which the same domestic control appears three times, so a better estimator raises numerator and denominator together and the rate is not simply the rate plus the improvement; recomputing every conditional and every robustness check inside a filtered specification is a different paper. What the result does establish is the direction and the size of what this paper's transparency costs: roughly two points of R^2 at the longest delay. That is the honest bound on the estimator limitation, and it is larger than several of the effects the paper argues over. The horizon evidence in Section 8.3 runs out at one month, where the rate has already fallen to 51% with an interval too wide to lean on; we do not know what happens at a quarter. Equity indices only. Close times are treated as standard, ignoring daylight saving, which moves them by an hour and never reorders the blocks. The index data are a retail export with no point-in-time guarantee, and one join in it is unverified: of the nine two-file exports, eight overlap and agree exactly on every overlapping row, while the S&P's two files abut without overlapping, so the only checks available there are that the join is contiguous and that no level shift appears across it. Section 3 gives both. That is the one series in the paper where a silent corruption at a single seam could not be detected by comparison, and it is the target series.

For VIX the same concern is now settled instead of flagged, because Cboe publishes its own daily history and we audited one against the other. On the 6,747 dates both carry, they are identical to the cent on 99.91% of rows and the mean absolute difference is 0.000222 index points; two rows fall outside this project's 0.5% seam tolerance, 28 November 2008 and 24 December 2012, both shortened sessions. The audit also found something we were not looking for. The retail export carries fifteen rows on dates United States equities were shut, 11 September 2001 among them: a fabricated observation rather than a discrepancy, and one no admissibility rule here could have caught, because the row is perfectly well formed. All fifteen fall outside the panel, so no model was ever shown one. VIX is Cboe's file throughout this paper; substituting it moved no cell of Section 8 by as much as a thousandth of R^2 . We report that as measurement instead of reassurance, and note that a retail export of any series can contain rows the exchange never produced.

The official file does not, however, supply the observation Section 3 actually wants, because it carries one open, high, low and close per day and no intraday snapshot at all. The 21:00 value of VIX existed and is the right regressor; it is not in any daily product, official or retail, and the lagged and same-day arms bracket it without recovering it. Differences at $\delta \leq 2$ fall inside the simultaneous bands. The variance-scale loss is dominated by a handful of crisis days, so its intervals are wide enough to be uninformative at short delays.

12. Reproducing this

Every number above is printed by one of thirty-three scripts: `lab03_crosssection.py`, `lab05_robustness.py`, `lab06_inference.py`, `lab07_estimation_cost.py`, `lab08_implied_vol.py`, `lab09_nonlinearity.py`, `lab10_loss_scale.py`, `lab11_markets_continuous.py`, `lab12_appendix.py`, `lab13_origin_median.py`, `lab14_appraisal_smoothing.py` and `lab15_nonlinear_given_iv.py` and `lab16_clark_west_shrinkage.py`, `lab17_horizons.py` and `lab18_economic_reading.py` and `lab19_vstox_third_series.py`, `lab21_stronger_inference.py` and `lab22_factor_benchmark.py` and `lab23_compressed_everywhere.py`, `lab27_regime_conditioning.py` and `lab28_orthogonal_breadth.py`, `lab34_data_snooping.py` and `lab35_purged_cv.py` and `lab37_lead_lag.py`, `lab38_domestic_baseline.py` and `lab39_temporal_stability.py` and

lab40_level_or_normaliser.py and lab41_conditional_anatomy.py and lab42_factor_model.py, lab43_official_vix.py and lab44_horizon_matched_iv.py and lab45_effective_age_interval.py and lab33_ragged_edge.py, all run on the data of Section 3 under a fixed seed. They are collected at github.com/SunnyAlexV/delayed-information-cross-section, together with the data, the reference output and both papers. A single command, `python run_all.py --check`, reruns every script and compares each line against that stored output. Four of those scripts were additionally reproduced on a second machine running a different operating system and matched digit for digit; the rest have been run only on the platform that generated the reference output, and we state that instead of the stronger claim. The seam audit prints on every run, including the seam it could not verify. Nothing is reported that the scripts do not produce.

A second command, `python verify_paper.py`, reads the source of both papers and checks every figure in them against that stored output. It fails on any disagreement.

13. Does a clean lag resemble a stale mark?

Section 14 will argue that what carries from this paper to an illiquid asset is the apparatus rather than the numbers. One part of that claim can be checked rather than hoped for, and this section checks it. Every result here models staleness as a clean lag: the forecaster sees the state exactly as it stood δ days ago and nothing since. A marked asset is not stale in that way. It is revised on a schedule and held flat in between, and the revision is a partial adjustment toward current evidence, not a full mark, which is how the property literature has modelled appraisal indices for thirty years [8]. If the substitution rate under that kind of staleness looked nothing like the rate under a clean lag, $R(\delta)$ would be a curve about clean lags alone.

It does not. Rebuilding the domestic block as a mark revised every q days and held between revisions, and matching on mean staleness, gives 64% against the clean lag's 60% at ten days, 71% against 70% at thirty-one, and 70% against 72% at ninety-four. Adding partial adjustment on top moves it a few points the other way, 63% where matching predicts 70%, so the equivalence is approximate and we claim no more than that. One thing we got wrong on the way is worth recording: our first attempt modelled appraisal as pure exponential smoothing, which is not staleness at all. A weighted average of the recent past still contains today's observation, so it costs almost nothing. R^2 falls from 0.499 to 0.456 across a mean lag of eleven weeks, against 0.499 to -0.035 under a clean lag, and leaves almost no damage for breadth to repair. Holding a number flat between revisions is what makes a mark stale, not blurring it.

14. What to do with this

Someone holding a stale mark on an asset with a liquid options market should buy the options data, and should buy the tenor that matches their forecast horizon, not the headline one. Section 8 provides strong evidence for the first, and the foreign cross-section adds nothing measurable once implied volatility is in hand. Section 8.5 provides it for the second: for a five-day question the nine-day index beats thirty-day VIX at every delay we test, by 0.024 to 0.041 of R^2 , which is a larger prize than most of what the rest of this paper argues over and costs nothing but choosing the right series.

Someone holding a stale mark on an asset with no options market has breadth as the only candidate on the table, and it is no corner case: it is the case Section 1 opens with. Quarterly-marked private equity, appraised property and funds reporting a lagged net asset value have no listed option chain

on the asset itself, so the substitute that wins Section 8 is not available to any of them. The redundancy result therefore narrows where the equity measurement transfers, not where the question matters. Sections 4 to 6 show how that worth can be measured and give a liquid-market benchmark for it, rather than an estimate for the illiquid asset itself: about 80% of the delay-induced loss recovered in stressed markets, where the estimate is stable and tight, 81% at eleven weeks with an interval of [74, 87], and an amount this sample cannot determine in calm ones. One practical point belongs with those numbers and is easy to miss, because the paper reports it as a robustness result. The rates above come from the prespecified seven-regressor model, which is not the model to run. Section 5.1 shows a single real-time factor beating all seven series at every delay and raising $R(55)$ to 75.2%, and an equal-weighted mean of the seven does nearly as well for no estimation at all. The seven coefficients cost one to five points of R^2 to fit and buy nothing back. Anyone building this should compress the cross-section first and treat the seven-regressor figures as the conservative benchmark they are. The unconditional figure is 72% against the paper's own control and 68% against the stronger control of Section 10.1; we quote the conditional numbers because they are the ones that mean anything. Against that sits a cost of one to five points of R^2 for carrying the extra series. One condition comes with those figures, and Section 9.3 makes it explicit: the foreign data has to be current. Every rate quoted in this paper assumes the cross-section arrives the same evening, and a feed one day behind turns Table 14's 36% into 9% at three days and its 51% into 29% at five. Before buying breadth, check that it will arrive on time. A cross-section older than the mark it is replacing is worth nothing, and at the shortest delays it is worth less than nothing.

One generalisation is worth refusing explicitly. It would be convenient to present Figure 1 as a benchmark for how quickly information decays in delayed systems, and it is not one. Those curves are a measurement of one index against seven specific peers over one sample. The correlation between a private portfolio and the public cross-section is a different quantity, and nothing here bounds it. A reader who wants a number for their own asset will not find it in this paper.

What does carry across is the apparatus rather than the numbers. $R(\delta)$ is defined in units that survive a change of loss function, market or candidate. The delay sweep is a design anyone with a delayed series and a candidate substitute can run, and the horse race in Section 8 is the question to ask first, because its answer decides whether breadth is worth buying at all. The cost decomposition in Section 5 separates what a source knows from what a forecaster keeps, the distinction that turned our own most confusing result into an ordinary one. Pointed at an appraisal-based or quarterly-marked series, that machinery would produce the numbers this paper cannot, and the check above says the delay model it carries is the right shape for that series, even though the numbers are not.

15. What would test this next

The universe here is eight equity indices, and the obvious way to widen it is the one this paper predicts will fail. Section 5.3 splits the foreign block into its leading real-time component and the six components orthogonal to it, and finds the entire recovery in the first: the remainder is negative at every delay and beats the stale domestic model at none. A ninth and a tenth equity index would supply more of the same factor and more coefficients to estimate, and Section 5 has already priced that trade. Breadth within an asset class is not the frontier.

Breadth ACROSS asset classes is, and it is a sharper test than it first appears because this paper has to make a prediction about it. Sovereign bond futures, the major commodities and the G10 currencies all

have observable daily ranges and therefore the same Yang-Zhang decision variable, so the design transfers with nothing changed but the panel. The stress ladder of Section 9.1 suggests they load on the same global risk factor, since cross-asset correlation is what converges in the episodes on that ladder. If they do, they will be redundant for the reason the orthogonal equity components are redundant, and the mechanism claimed here will have survived a considerably harder test than another equity index.

The way it breaks is the more interesting outcome, and it is specific enough to look for. Commodity volatility has drivers that no equity cross-section can see: an inventory report, a harvest, a production decision by a cartel. If a commodity block adds where equity breadth does not, then “a global volatility factor seen early” is too coarse a description of what is happening, and the right cross-section depends on what the stale asset IS, and is not one panel for everyone. A farmland book and a private credit book would then need different peers, which is a materially different recommendation from the one Section 14 gives and would be worth the trouble of discovering.

Foreign exchange should be run first, for a reason Section 9.3 makes plain, and for no reason about currencies. FX trades continuously across the week, so it has no admissible-timestamp problem at all: there is no previous session for a currency pair, no peer that closes after the target, and none of the shift rules in Section 3. That also means it is exempt from the penalty in Table 14, the largest practical threat this paper has to its own recommendation. A substitute that is current by construction is precisely what a holder of a stale mark wants, and of the candidates listed here FX is the only one that is current by construction, not by luck of the clock.

Two warnings for whoever runs it. The models are the easy part; the data hygiene is not. Futures require contract rolls, so a continuous series carries a seam every quarter instead of the single seam per export audited in Section 3, and the audit, not the estimation, is where that work will actually go. And the admissibility table stops being a list of closing times: a bond future and its underlying cash market can trade on different calendars, so the rule that a peer counts only when its timestamp precedes the target's needs restating instrument by instrument instead of country by country.

Three other threads are open and named where they arise: the illiquid-asset target this whole exercise is about and cannot reach, in Section 11; whether the rate keeps rising past one month, where Section 8.3's evidence runs out with an interval too wide to lean on; and, in the companion note [13], ranking forecasters against a tail target, which needs more independent volatility spikes than nine years of one index contains. The cross-asset panel proposed above would help with the last of those as a side effect, since spikes accumulate across markets as well as through time.

Disclosure of AI use

Claude (Anthropic) was used throughout this work: to write the replication and execution scripts, to draft and revise the text, and to assist in the iterative design of several tests. All research questions, design decisions, and claims were directed and accepted by the author, who is solely responsible for the contents. Every number reported is produced by the scripts in the public repository and checked against them by `verify_paper.py`.

Appendix. A confidence set for a ratio whose denominator can vanish

The substitution rate is a ratio, and Section 7 reports that inverting a HAC t-test after Fieller leaves every published interval standing. This is how that was computed, and why it was not the first answer we got. Fieller's theorem inverts a t-test on the linear combination $\text{num} - R\text{-den}$, which is zero at the true rate, and solves the resulting quadratic for the set of rates not rejected. When the denominator is well separated from zero the solution is an ordinary interval close to the delta-method one. When it is not, the quadratic's leading coefficient changes sign and the confidence set is correctly a union of two half-lines, or the whole line: an honest statement that the data do not pin the ratio down, and one a percentile bootstrap cannot make. We have met the symptom in this project without naming it, in an interval of $[-154\%, +50\%]$ at one point, and treated it as a nuisance when it is an identification problem. Two things about the calculation are worth recording. The variances must be HAC, with the same Bartlett kernel used in Section 7, because the five-day targets overlap: the first version used iid variances and returned intervals *narrower* than the block bootstrap at every delay, which is impossible for overlapping targets and was the only sign that anything was wrong. And $\delta = 0$ is absent from the comparison throughout, because there the denominator is identically zero by construction, the one delay at which the ratio does not exist at all, rather than being merely hard to pin down.

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